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MIL-STD-361-3(TM)
15 JULY 1992

SUPERSEDING
(See 6.5.)

MILITARY STANDARD

MANUALS, TECHNICAL:

REQUIREMENTS FOR GRAPHICS

AMSC A6752

AREA TMSS

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FOREWORD

1. This military standard is approved for use by the Department of the Army and is available for use by all Departments and Agencies of the Department of Defense (DOD).

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, USAMC Materiel Readiness Support Activity, ATTN: AMXMD-EP, Lexington, Kentucky 40511-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

3. This document is one part of an 11-part bookform standard to be used in the preparation and acquisition of technical manuals that are essential to support (operate and maintain) the various types of equipment and weapon systems within the Department of the Army.

4. This document covers the graphics requirements for the preparation of technical manuals.

5. Other parts of this multipart standard are:

MIL-STD-361-1	-	Writing Style and Format Requirements.
MIL-STD-361-2	-	Comprehensibility Requirements.
MIL-STD-361-4	-	Quality Assurance Requirements.
MIL-STD-361-5	-	Requirements for Packaging.
MIL-STD-361-6	-	Introductory Information with Theory of Operations Requirements.
MIL-STD-361-7	-	Operator and Preventive Maintenance Instructions.
MIL-STD-361-8	-	Maintenance Instructions.
MIL-STD-361-9	-	Troubleshooting Procedures.
MIL-STD-361-10	-	Lubrication Instructions.
MIL-STD-361-11	-	Repair Parts and Special Tools List (RPSTL).

6. By incorporating standardized and service-unique technical manual requirements, this document and its dash-numbered companion parts, along with the general acquisition specification, MIL-M-63008, Acquisition/Preparation Requirements, support the acquisition of technical manuals.

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1. SCOPE

1.1 Scope. This document establishes the content and procedural requirements for preparation of graphics presentations for technical manuals (TMs), revisions, supplements, and changes.

1.2 Purpose. This graphics standard supplements the various requirements in the other parts to this 11-part standard for specific types of technical manuals and related publications and does not alone deliver any technical data. (NOTE: The figures used in this document are examples only. The text of this document takes precedence over the figures.)

1.3 Applicability. The applicability of the requirements contained in this document is governed by the equipment, systems, or weapon systems for which the technical manuals are being developed. This document is applicable for use by the Department of the Army.

1.4 Selective application and tailoring. This document contains some requirements that may not be applicable to the preparation of all technical manuals. Selective application and tailoring of requirements contained in this document are the responsibility of the contracting activity and shall be accomplished through the use of MIL-M-63008. The applicability of some requirements is also designated by one of the following statements:

- unless requirement is specifically excluded by the contracting activity; or
- as/when specified by the contracting activity.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation. (See 6.2.)

SPECIFICATIONS

MILITARY

MIL-D-1000	-	Drawings, Engineering and Associated Lists.
MIL-M-63008	-	Manuals, Technical: Acquisition/ Preparation Requirements.
MIL-P-38790	-	Printing Production of Technical Manuals: General Requirements for.

STANDARDS

MILITARY

MIL-STD-12	-	Abbreviations for Use on Drawings, and in Specifications, Standards, and Technical Documents.
MIL-STD-17-1	-	Mechanical Symbols (Other Than Aeronautical, Aerospacecraft, and Spacecraft).
MIL-STD-17-2	-	Mechanical Symbols for Aeronautical, Aerospacecraft, and Spacecraft Use.
DOD-STD-100	-	Engineering Drawing Practices.
MIL-STD-361-1	-	Manuals, Technical: Writing Style and Format Requirements.
MIL-STD-361-2	-	Manuals, Technical: Comprehensibility Requirements.
MIL-STD-361-4	-	Manuals, Technical: Quality Assurance Requirements.
MIL-STD-361-5	-	Manuals, Technical: Requirements for Packaging.

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MIL-STD-361-9	-	Manuals, Technical: Troubleshooting Procedures.
MIL-STD-361-11	-	Manuals, Technical: Repair Parts and Special Tools List (RPSTL).
DOD-STD-863	-	Wiring Data and System Schematic Diagram, Preparation of.

(Unless otherwise indicated, copies of federal and military specifications and standards are available from Standardization Documents Order Desk, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5904.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents that are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are issues of the documents cited in the solicitation. (See 6.2.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/IEEE STD 91-1984	-	Graphic Symbols for Logic Functions.
ANSI/IEEE STD 200-1975 (R1989)	-	Reference Designation for Electrical and Electronics Parts and Equipments.
ANSI/IEEE STD 260-1978 (R1985)	-	IEEE Standard Letter Symbols for Units of Measurement.
ANSI/IEEE STD 280-1985	-	Letter Symbols for Quantities Used in Electrical Science and Electrical Engineering.
ANSI/IEEE STD 315-1975 (R1989)	-	Graphic Symbols for Electrical and Electronic Diagrams Including Reference Designation Class Designation Letters.
ANSI Y14.15-1966 (R1973)	-	Electrical and Electronics Diagrams.
ANSI Y14.17	-	Fluid Power Diagrams.
ANSI Y32.10	-	Diagrams, Fluid Power, Graphic Symbols For.

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(Application for copies should be addressed to the American National Standards Institute Inc, 1430 Broadway, New York, New York 10018-3308.)

2.3 Order of precedence. In the event of a conflict between the text of this document ("This document" includes all volumes of MIL-STD-361 series.) and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. DEFINITIONS

3.1 Acronyms used in this standard. The following acronyms listed in this standard are defined as follows:

a. ANSI - American National Standards Institute.

b. DOD - Department of Defense.

c. DODISS - Department of Defense Index of Specifications and Standards.

d. FRC - Final Reproducible Copy.

e. IEEE - Institute of Electrical and Electronics Engineers, Inc.

f. RPSTL - Repair Parts and Special Tools List.

g. TM - Technical Manual.

3.2 Block diagram. A modified schematic diagram in which each group of maintenance-significant components that together perform one or more functions is represented by a single symbol or block. The block or symbol representing the group of components shows simplified relevant input and output signals pertinent to the subject diagram.

3.3 Callout. Anything placed on an illustration to aid in identifying the objects being illustrated, such as index numbers, nomenclature, leader lines, and arrows.

3.4 Clutter. A condition in which the amount of information in a given area is so great that confusion and errors of interpretation occur.

3.5 Comprehensibility. The completeness with which a user in the target audience understands the information in the TM.

3.6 Continuous tone photographs or drawings. Continuous tone photographs or drawings have a continuous gradation of tonal values ranging from light (white) to dark (black), including gray. These tonal values are not created by lines or dots. (See 3.9.)

3.7 Functional diagram. A type of illustration in which symbols are connected by lines to show relationships among the symbols. The symbols may be rectangles or other shapes, standard symbols representing components or functions, or pictorials representing equipment or components. The lines may represent procedures or processes, such as signal or logic flow, and physical items, such as wires. Functional diagram includes schematics, wiring and piping diagrams, logic diagrams, flow charts, and block diagrams.

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3.8 Graphic(s). Any type of presentation or representation which gives a clear visual impression.

3.9 Halftones. Halftones are the tonal values of gray and black created by lines or dots. It is a conversion of a continuous tone print. (See 3.6.)

3.10 Illustration. A general term meaning graphic presentations of all types. Illustrations include pictorials, functional diagrams, and line graphs. This term is used instead of such terms as figure, graphic, drawing, diagram, and artwork.

3.11 Index number. Numbers used on an illustration, legend, or in text to refer to an object.

3.12 Nomenclature (equipment). The approved name or alphanumeric identifier assigned to an item, equipment, or component in agreement with an organized designation system.

3.13 Phantom. Portraying an item (i.e., part, equipment, etc.) on an illustration with broken lines rather than solid lines to show the item as subdued.

3.14 Pictorial. A type of illustration showing the physical appearance of equipment or component parts. This term is used instead of such general terms as illustration, drawing, and diagram.

3.15 Prescreening. A process in which a clear material with a dot pattern or crossing opaque lines is used through which an image is photographed in making a halftone.

3.16 Procedure. The sequence of actions that make up a task.

3.17 Readability. Text comprehensibility measured by such variables as number of syllables, words, and sentences.

3.18 Schematic diagram. A graphic representation showing the interrelationship of each component or group of components in the equipment. The essential characteristic of these diagrams is that every maintenance-significant functional component is separately represented.

3.19 Steps. Actions, proceedings, or measures occurring in a series or logical sequence.

3.20 Tasks, user. A sequence of user actions with a beginning and an end. User tasks relate to installation, checkout, operation, and maintenance of systems or equipment.

3.21 Technical manuals (TMs). Manuals that contain instructions for the installation, operation, maintenance, and support of weapon systems, weapon system components, and support equipment. TM information may be presented, according to prior agreement between the contractor and the Government, in any form or characteristic, including hard printed copy, audio and visual

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displays, electronic embedded media, disks, other electronic devices, or other media. They normally include operational and maintenance instructions, parts lists, and related technical information or procedures exclusive of administrative procedures.

3.22 Text. The written parts of the technical sections excluding labels, legends, and callouts in illustrations.

3.23 User. A person supported by the technical manual in performing technical tasks.

3.24 Wiring diagram. Diagram illustrating signal flow or wiring connections.

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4. GENERAL REQUIREMENTS

4.1 General illustration requirements. Graphic presentations developed for technical manuals (TMs) shall conform to this document and MIL-STD-361-1.

4.1.1 Illustration detail.

a. Style and techniques shall be of a quality that will produce illustrations which will clearly, adequately, and economically portray the information to be illustrated. The amount of detail should be limited to that required to support the content of the illustration. Unnecessary detail should be avoided. Illustrations shall be integrated with the text and shall be used when text alone cannot adequately communicate the message. Illustrations shall be used to describe an item, process, or procedure; clarify text; present information difficult to describe by text alone; call attention to details; and provide identification of assemblies, parts, and tools, etc. When text alone is not adequate, illustrations shall be used for depicting procedures such as disassembly, assembly, removal, and installation. Illustration views shall be presented as the user would view the item in the performance of the associated task.

b. Illustrations other than foldouts shall be located as near as possible to the point at which they are first referenced in the text, except where this would require unnecessary duplication of illustrations. Full-page illustrations, placed sideways on a page, shall be turned 90 degrees counterclockwise. The figure number and title for a turned illustration shall be placed at the bottom of the page with the TM in its normal position. When the majority of illustrations are turned, the figure number and title shall also be turned to appear below the illustration. (See figure 1.)

4.1.2 Scale. Illustrations shall be prepared to as small a scale as possible to be consistent with effective use of space and with all essential detail legible. Illustrations shall be the same size as the areas they will occupy on TM pages. See MIL-STD-361-1 regarding foldout pages.

4.1.3 Letter size. The scale of text on illustrations shall provide for a minimum final letter size, when printed, of 8 points. (See also MIL-STD-361-2.)

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5. DETAILED REQUIREMENTS

5.1 Types of illustrations.

5.1.1 Photographs versus line drawings. Unless requirement is specifically excluded by the contracting activity, only line drawings shall be used. However, existing artwork shall not be reworked solely to remove detail and to convert continuous tone art and photographs to line drawings. As specified by the contracting activity, existing illustrations, photographs, or engineering drawings may be used where they meet the requirements of this document. Line drawings shall be prepared from source data and equipment photographs. When a photographic line tracing is prepared, it shall be of high reproduction quality.

5.1.2 Photographs. When approved for use by the contracting activity (5.1.1), photographs shall be detailed and sharp; free of heavy shadows, distorted objects, and cluttered foregrounds or backgrounds; and give good contrast from white, middle tones, and black.

5.1.2.1 Prescreened photographs. When approved for use by the contracting activity (5.1.1), prescreened photographs are acceptable as reproducible copy provided they will not be rescreened and are of proper quality, size, and mounted on the reproducible copy of the text page. When prescreened photographs are used, they shall be clearly marked to indicate prescreening. As specified by the contracting activity, unscreened continuous tone photographs and/or original illustrations shall be supplied with reproducible copy.

5.1.2.2 Continuous tone artwork. Continuous tone artwork, whether photographs or drawings, shall be clear in detail, sharp in contrast of tones, and have light and shadow in proper relation to a consistent light source. The background shall be light enough to contrast fully with the subject photographed and shall extend the full width and depth of the photograph or drawing.

5.1.3 Line drawings. Line drawings shall be of high reproduction quality. A suitable material, capable of maintaining consistent and permanent high-density reproducible values (such as acetate, or equivalent), shall be used for preparing the line drawings.

5.1.4 Engineering drawings. Unless requirement is specifically excluded by the contracting activity, engineering drawings shall not be used as illustrations. When used, they shall be in accordance with DOD-STD-100 and shall be modified, as necessary, to meet the content, style, arrangement, legibility, format, and production requirements described in this document and the contract. All unnecessary data that would reduce the comprehension or clarity of the illustration shall be removed. They must be reduced or redrawn to meet page size restrictions.

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5.1.5 Combination illustrations. Unless requirement is specifically excluded by the contracting activity, combining photographs or continuous tone artwork with line drawings shall not be used. When approved, presentation of a subject by combining photographs or continuous tone artwork with line drawings shall be limited to where this treatment presents the subject more accurately or more clearly.

5.1.6 Multisection or multiview illustrations.

a. Each section of a multisection illustration shall be identified by a capital letter. Sections may or may not be titled, but if one section is titled, all shall be titled. Each title, with the identifying letter as its first character, shall be centered with respect to the section to which it applies. Where titles are not used, the identifying letters shall be centered. Identifying letters and titles shall be larger and bolder than any other lettering on the illustration. Sections shall be clearly separated by lines. Separation by shading shall not be used. (See figure 2.)

b. Multiviews shall be prepared when necessary to identify significant features on an illustration. Each view shall be oriented and enlarged as necessary to identify significant features of the item. Orientation shall be by use of arrows or by text (i.e., "Rotated 180 degrees.") as it relates to the main illustration. Multiviews shall be presented in the same manner as multisection illustrations.

5.1.7 Cartoons. Unless requirement is specifically excluded by the contracting activity, the use of cartoon type drawings and other similar visual techniques shall not be used. When used, such drawings shall not include copyrighted cartoon characters and must serve a functional purpose.

5.2 Color in illustrations. Color of ink specified to be used in printing shall be limited to black and shades of black (one color), unless prior approval is provided by the contracting activity from the U.S. Army Publications and Printing Command. Written approval, designating additional color(s) to be used if deemed necessary, shall be provided by the contracting activity. When color (other than black) is required, it shall be held to the minimum absolutely necessary to highlight or clarify important information. The number of colors shall be kept to a minimum by use of various techniques such as tints, patterns, cross-hatching, and dots. Any number of shades of a primary color used shall be considered as one color (e.g., a two-color printing could consist of black and three shades of red). When color is approved/specified, the primary colors of red and blue shall be used first. Yellow shall not be used alone.

5.2.1 Separation of colors. The black and white, or key, illustrations shall be mounted on an artboard or prepared on a material such as acetate. Each succeeding color overlay shall be

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made on acetate, or equivalent, and attached securely to the artboard or key acetate. Each color overlay shall be marked with its correct color. (See also 5.4.2 and MIL-P-38790.)

5.3 Elements of illustrations.

5.3.1 Border rules and boxes. Unless requirement is specifically excluded by the contracting activity, border rules and boxes shall not be used for single illustrations, but shall be used to separate multisection illustrations on the same page or detail/locator views. (See 5.1.6 and figures 2 and 3.)

5.3.2 Use of the human figure. When necessary, illustrations may include a human figure or parts of the body. Jewelry shall not appear in any illustration. The illustrated human figure shall not obscure necessary details of the item(s) being illustrated. The clothing for the human figure shall be illustrated according to samples provided by the contracting activity. A cross-section of races and sexes shall be used except when a task is to be performed by male personnel only.

5.3.3 Credit lines. The photographer's or illustrator's name shall not appear on any illustration; neither shall a manufacturer's name, symbol, or trademark appear on illustrations for the purpose of identifying the illustration. When a contractor's illustration identification number is approved for use (See MIL-STD 361-1.), it shall be in approximately 4- to 6-point type placed in the lower right-hand corner of the illustration sufficiently removed to avoid being confused as part of the illustration.

5.3.4 Callouts. Index numbers, reference designations, nomenclature, leader lines, legends, and other identifiers shall be used, when necessary, to identify significant features. Callouts shall be prepared by a mechanical or electronic method, rather than by freehand lettering. (Engineering drawings prepared in accordance with 5.1.4 are acceptable.) Unless requirement is specifically excluded by the contracting activity, type size shall be no smaller than 8 points and no larger than 10 points. Lettering shall be in uppercase. Nomenclature shall appear on illustrations only if it can be done without crowding or reducing type size so as to make reading difficult. (Diagram callouts shall be no smaller than 8 points.) Callouts shall be placed in the background areas of illustrations when practical.

5.3.4.1 Index numbers.

a. Titled figures. Index numbers for each titled figure shall start with Arabic numeral 1 and continue consecutively. (All the multisheet illustrations in a set shall be considered one figure.) As specified by the contracting activity, index numbers used for any procedure (operation, assembly, disassembly, remove, install, repair, etc.) shall be in clockwise sequence (beginning at 11:00 o'clock), disassembly order, or in order of mention in the text. All items shown as exploded shall be

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identified (figure 3) or shall be shown in phantom (figure 4).

b. Untitled procedural illustrations. Index numbers for step-by-step, untitled, procedural illustrations shall start with Arabic numeral 1 and continue consecutively. As specified by the contracting activity, index numbers shall be in disassembly sequence, clockwise sequence (beginning at 11:00 o'clock), or in order of mention in the text. (See figure 5.)

c. RPSTL and DMWR figures. For RPSTL and DMWR figures the sequence of index numbers shall start at 11:00 o'clock and proceed clockwise. (See figure 6.)

d. Added index numbers. When an illustration is changed (except RPSTL), index numbers added between existing numbers shall be the same as the preceding index number with an added decimal number (e.g., 22.1, 22.2). This system shall also be used in basic manuals when errors are discovered so late in preparation that renumbering of all following index numbers would delay submittal. Index numbers with added decimal numbers shall be eliminated for a revision.

5.3.4.2 Nomenclature. Nomenclature of more than one line shall have the left margin justified when placed on the right side of the illustration, right margin justified when placed on the left side, or shall be stacked with the text centered when on the top or bottom of the illustration. Nomenclature on illustrations shall be in all capital letters. All lines of copy shall parallel the horizontal edges of the figure, whenever possible. (See figure 7.)

5.3.4.3 Leader lines and arrowheads. Leader lines or arrowheads shall end close to the callout and object; or touch, but not enter, the objects to which they apply. Lines shall be uniform, short, and as straight as possible; dogleg-shaped lines shall be used only as necessary. Lines and arrowheads shall not cross or come in contact with other callout lines or arrowheads nor shall they obscure essential details. Leader lines should be placed at an angle. Leader lines and arrowheads shall be highlighted or changed from black to white to make them easier to follow. Arrowheads may be added for clarity. Arrowheads shall be uniform in shape and size when multiple arrowheads are used on a page. (See figure 7.)

5.3.5 Callouts on overlays. The callouts for halftones and continuous tone photographs shall be placed directly on the halftone or photograph or on transparent overlays. When overlays are used, registration marks in accordance with 5.4.2 shall be added. (See figure 8.)

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5.4 Preparation of illustrations for printing.

5.4.1 Crop and size marks. Each separately supplied, manually created illustration shall have the reproduction area defined by crop marks appearing on each of the four corners marking the horizontal and vertical dimensions of the area to be reproduced. The lines shall extend no closer than 1/4 inch to the outside of the reproduction area. The exact reproduction size shall be indicated between crop marks. Marks shall not be drawn with ballpoint pen or grease crayon. Crop marks shall be approximately 3/8 inch long and shall not cross or touch. (See figure 9.)

5.4.2 Registration marks. The basic artboard and each overlay shall contain a minimum of three registration marks located outside the reproduction area, one centered on each side and one centered at the top or bottom. The registration marks on each overlay shall accurately align with one another and with the basic artboard. (See figure 8.)

5.4.3 Mounting of illustrations. Illustrations which are to be incorporated with text manually (including continuous tone photographs) shall be securely fastened to an artboard by a process that will protect the illustration and/or text, will not discolor or distort, and will ensure there are no wrinkles and blisters. As specified by the contracting activity, illustrations and text prepared together electronically or by photographic method shall be mounted on an artboard, or final reproducible copy shall be the original electronic or photographic presentation. When mounted on an artboard, the size of the artboard shall allow for a minimum border of 1-1/2 inches on all four sides of the image area to accommodate required external identification markings. Security classification, if applicable, shall be placed at the top and bottom center of the reverse side of the artboard, outside the image area on the front of the artboard, and outside the image area on each overlay, if any. (See figure 8.)

5.4.4 Identifying illustrations. Illustrations not affixed to the completed final reproducible copy (FRC) shall bear the TM number and figure number outside the reproduction area. The security classification, if applicable, shall be placed at the top and bottom center outside the reproduction area. The word "TOP" shall be placed at the top of the artboard or the electronic or photographic presentation. (See figure 8.)

5.4.5 Covering the artboard. All board-mounted text and illustrations shall be protected by an inner flap of non-oil tissue or vellum and an outer flap of heavy paper. The tissue overlay and protective outer flap shall be secured to the back of the artboard at the top and folded over the face of the artboard. The figure number (if applicable), drawing or illustration number, page number, and TM number shall appear in the outside upper right corner of the outer flap. The security classification, if applicable, shall also be placed at the top and bottom center of the outer flap. (See figure 8.)

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Electronically or photographically generated pages do not require an inner flap of non-oil tissue or vellum.

5.5 Illustration preparation.

5.5.1 Illustration style and format preparation. Illustrations shall be planned, laid out, scaled to a size to effectively portray the required details, and prepared to properly present information. The requirements of 5.1 through 5.5 for preparation and identification, including numbering, shall be followed. (See also MIL-STD-361-1.) The comprehensibility and readability shall be developed in accordance with MIL-STD-361-2. Each illustration should contain only that information directly relevant to a specific user task. However, if one illustration will support two or more requirements, the illustration may be repeated. For example, when the illustration meets both requirements, illustrations prepared to support operating should be used to support maintenance procedures.

5.5.2 Placement of illustrations and related text. Where possible, text and related illustrations shall be arranged so they can be viewed at the same time. If this is not possible (for example, more than one full-page illustration), the illustrations shall be placed as soon after the first reference in the text as possible.

5.5.3 Line drawing details. The darkness and sharpness of lines shall be sufficient to reproduce clearly at required reproduction size without additional treatment. (See figure 10.) Secondary lines, such as those used to indicate extensions or measurements, shall be lighter but strong enough to reproduce clearly at reproduction size. Shading may be used to give substance and form to the item depicted, to sharpen the contrast between the subject and its background, or to increase effectiveness. Shading and shadows shall be used only when necessary to provide a clear understanding of form, shape, or depth. Shading effects shall not be used for decorative purposes. Accented lines may be used to emphasize detail when necessary. Lined, cross-hatching, or mechanical patterns used instead of color shall remain clearly defined on the final reproducible copy. (See figure 11.)

5.5.4 Illustration callouts and identifiers. Callouts or identifier numbers on illustrations shall have a leader line or sweep arrow connecting them to the correct point on the illustration. (See figure 7.) Leader lines or sweep arrows shall be used to help the readers orient themselves with respect to the illustration and to provide directional movement in tasks. The callouts or identifier numbers shall be limited to those used in the accompanying text and shall be keyed to the text. When an item is first illustrated and its location has not yet been specified, a "simplified" general locator illustration may be used to identify the location of the equipment item within the system. (See figure 7.) In addition to the normal situations described above, the following describes special situations:

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a. Two tasks using the same illustration shall be included on the same page if space permits and the tasks are in sequence. (See figure 5.)

b. If an illustration occupies less than a full page, associated text shall appear with the illustration to the extent possible without crowding. (See figure 5.)

c. To prevent the use of a foldout page, illustrations shall be placed sideways on the page (rotated 90 degrees counterclockwise). (See figure 1.)

d. Illustrations shall be placed on the same or facing page of associated text when practical.

e. Full-page illustrations shall not be repeated unless necessary to support multipage descriptions of tasks or to support a different requirement in another part of the TM, as appropriate. Whenever practical, they shall be numbered, titled, and referenced rather than repeated.

5.5.5 Final reproducible copy leading and vertical spacing. Layout shall conserve space without lessening usability or clarity of material. Blank pages and spaces shall be avoided, whenever possible. Leading and vertical spacing shall be used for best legibility and conservation of space. Layout practices shall not result in:

a. The first line of a paragraph being at the bottom of a page or column.

b. The last line of a paragraph being at the top of a new page.

c. A paragraph title falling on the last line of a page or column.

d. Warnings, cautions, and notes being divided so that first lines or group of icons appear on one page and remaining lines or group of icons on another. (First lines or group of icons may appear in the left column with remaining lines in the right column on the same page.)

e. Warnings, cautions, and notes being separated from the paragraph to which they apply. (Warnings, cautions, and notes may appear in the left column with applicable paragraphs in the right column on the same page.)

f. Undesirable location of an illustration or table.

5.6 Change symbols.

5.6.1 Change symbols for figure titles. When anything within the figure has been added, deleted, or changed, a 6-point (3/32-inch) wide vertical line shall be placed to the left of the figure number and title. When a figure is deleted or added, the

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change symbol shall also be placed to the left of the number. (In addition, for deletions the title is to be replaced with a statement such as: Figure 2-1 DELETED.)

5.6.2 Change symbols for illustrations. (See also MIL-STD-361-1 for general information on changes and MIL-STD-361-11 for preparing changes to RPSTLs.) Unless requirement is specifically excluded by the contracting activity, a miniature pointing hand shall be used for illustrations (other than diagrams and schematics) to highlight the area containing the changed information. Changes confined to the same general area shall be indicated only once on the illustration. A vertical line next to changed text and callouts on illustrations may be used in lieu of a pointing hand. If a callout is deleted from an illustration, the word DELETED shall be placed after the appropriate number in the legend, if applicable. A vertical line next to changed material may be used on a chart or graph. As specified by the contracting activity, screens (shading), screened (shaded) boxes, or miniature pointing hands shall be used to highlight changed areas of diagrams and schematics. (See figure 12.)

5.7 Schematic and functional diagrams.

5.7.1 Principles. In order to create properly drawn and organized schematic and functional diagrams, 1) a determination must be made as to what diagrams are needed; 2) the proper form (or combination of forms) must be chosen; 3) there must be an understanding of the methods of presentation, special content, and equipment coverage of each form; 4) the form that best fulfills the intended purpose must be chosen; and 5) the general methods and specific methods and requirements for the chosen form (or combination) must be followed. (For troubleshooting diagrams, see MIL-STD-361-9.)

5.7.2 Specification requirements. Use the following specifications as applicable:

Subject	Equipment Covered	Specification
Abbreviations	All	MIL-STD-12
Drafting Practices	Mechanical	
	Electrical and electronic	ANSI Y14.15-1966 (R1973)
	Fluid power	ANSI Y14.17
Engineering Drawings	All	DOD-STD-100, MIL-D-1000
Graphic Symbols	Electrical and electronic	ANSI/IEEE STD 315-1975 (R1989), ANSI/IEEE 280-1985
	Mechanical	MIL-STD-17-1 and -2
	Digital (Logic)	ANSI/IEEE 91-1984
	Fluid power	ANSI Y32.10

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Reference Designators	Electrical and electronic	ANSI/IEEE STD 200-1975 (R1989)
Unit Symbols	All	ANSI/IEEE STD 260-1978 (R1985)

5.7.2.1 Designations, diagrams, and symbols. Designations, diagrams, graphic symbols, and letter symbols shall be used on illustrations in accordance with specifications in 5.7.2. As specified by the contracting activity, new designations, diagrams, and symbols not covered by the specifications may be used if explained in the TM's How To Use This Manual section. Symbols shall be spelled out when the symbols cannot be reproduced by the equipment used to prepare the final reproducible copy (e.g., plus for +).

5.7.3 General methods. The specifications listed in 5.7.2 shall be followed for general methods in preparing diagrams. Other important requirements are:

a. Layout - Layout most diagrams, order signals, and components functionally. (For example, don't show signals in connection pin number order, but in functional order.) Avoid clutter - no more than 3 components per square inch or 20 crossed lines per 4 square inches. Allow 1/8 inch between parallel lines. Use a left-to-right functional flow with right-to-left feedback.

b. Consistency - Use a standard referencing system for associated text, signal flow, and other diagrams. Use standard graphic symbols when possible; if special graphic symbols are required, make them visually distinctive from other graphic symbols used and include them in a special symbols chart. Have standard names for hardware, functions, signals, controls, indicators, and switches, etc.

c. Appropriate detail - Include all information required to fulfill intended purpose of diagram; be careful not to overcrowd. Provide complete detail for hardware, function, signal identification, measurement data (voltages and waveforms), explanatory text, connectors, terminal boards, pin numbers, signal names, reference designators, component values and tolerances, and replacement components, etc. Clearly label all inputs and outputs and place them near edges of diagrams.

d. Inappropriate data - Leave out data not related to purpose of diagram. Include pertinent detail to ensure that details of nonreparable and nonreplaceable components are minimized.

5.7.4 Portraying signal flow. Signal flow, especially for electrical and electronic equipment, critically affects the

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understandability of diagrams. As applicable, use the methods for portraying signal flow in 5.7.4.1 through 5.7.4.3.

5.7.4.1 Signal connections. Three methods are available to represent signal connections: point-to-point method, highway method, and interrupted flow method.

a. The point-to-point method shall show each signal separately with a continuous line to represent its flow (figure 13).

b. The highway method shall blend two or more signals together in a single line (figure 13). This method is useful in showing the flow of a group of related signals. Any number of signals can be blended together. Any signal that has been blended into the main line shall be blended out at some other point on the line. Once a signal has been blended out of a line, it shall no longer be present on that line. Each signal blended in or blended out of the line shall be identified.

c. The interrupted flow method shall use special symbols to interrupt signal flow. This method may be used within a single sheet of a diagram, between sheets of a diagram, or between diagrams.

5.7.4.1.1 Techniques within a single sheet of a diagram. Three techniques are available for using the interrupted flow within a single sheet of a diagram:

a. Oval connector - Use to continue signals from one area of a sheet to another area. Any number of signals can be bracketed together. Each signal shall be identified at its source bracket and destination bracket. Oval connectors shall have a unique letter identifier inside the oval. (See figure 14.) The position of the source and destination connectors shall be identified by zone numbers (5.7.4.1.2b).

b. Signal returns - Use to continue signal returns within a single sheet of a diagram. Returns shall have a unique number identifier inside the network. Each return shall be labeled the first time it appears on the diagram (preferably on the left edge of the diagram). (See figure 15.)

c. Breakoff symbols - Only power forms, clock pulses, and other multiuse, minor symbols shall use the breakoff symbol technique. Each signal shall be identified adjacent to its breakoff symbols. The source of signals shall be shown at the left edge of the diagram. (See figure 15.)

5.7.4.1.2 Techniques between sheets of a diagram. Two techniques are available for using the interrupted flow between sheets of a diagram:

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a. Boat symbol - Use to continue signals from the right edge of one sheet to the left edge of the following sheet within a multisheet diagram (adjacent sheets of a diagram only). Use for single signals only. Boat symbols shall have a unique letter inside the boat. (See figure 16.)

b. Oval connector - Use to continue signals from one area of a diagram to another. Application is the same as within a single sheet of a diagram (figure 14). For identification of source and destination areas, use the following recommended zoning rules for multisheet diagrams: 1) Number vertical zones; letter horizonal zones. 2) Limit the number of horizontal zones to 10. 3) Always number zones as below, even if all zones are not used on any sheet.

Sheet 1 Starts with Zone 1
Sheet 2 Starts with Zone 11
Sheet 3 Starts with Zone 21, etc.

5.7.4.1.3 Techniques between diagrams. Three techniques are available for using the interrupted flow between diagrams:

a. Block technique - Include figure number (name optional), connector and pin numbers, and zone numbers (figure 16).

b. Oval connectors - Insert source and destination figure numbers before zone references (figure 14).

c. Pyramid diagram - Include diagram number from one diagram to another. (For example, include reference to I on diagram 2 and reference to II on diagram 1. See figure 16.)

5.7.4.2 Signal difference. Various techniques are available to indicate signal flow, signal importance, and type, such as line width (Use wide lines to represent major signals.), arrowheads (Use special arrowheads to indicate signal types.), and color (if approved by the contracting activity, see 5.2). (See figure 17.)

5.7.4.3 Signal junctions. The relative importance of signals can also be indicated by the way signal junctions are represented. Subordinate junctions shall be used to indicate differences in signal importance. Coordinate junctions shall be used to indicate equality in signal importance. (See figure 18.)

5.7.5 Conventional schematic and functional diagrams.

5.7.5.1 Schematic diagrams. Schematic diagrams shall show every maintenance significant functional component and functionalize complex assemblies which are nonreparable. Piece part detail shall be shown only when replacement is authorized at the maintenance level covered or when understanding is required for fault isolation. For nonreparable assemblies, all inputs and outputs shall be shown with enough detail to understand how

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inputs relate to outputs (complete details for simple circuits and symbols for complex circuits). (See figure 19.)

5.7.5.2 Functional diagrams. (See figure 20.)

a. Functional diagrams (block functional only) shall show the complete system or subsystem on one sheet (if possible). Methods to be used include functionalizing components; grouping subfunctions into functions; or continuing until the complete system or subsystem can be shown on one sheet (may be a foldout, when approved by the contracting activity, see MIL-STD-361-1).

b. Functional diagrams shall provide enough details to relate the input to output signals by using arrowheads to indicate signal flow direction when necessary and specifying signal characteristics and tolerances in pictorial or tabular form.

c. Functional diagrams shall account for all maintenance significant components by ensuring the user can relate the schematic diagram to the functional diagram; blocking the components on the schematic to correspond with blocks on the functional; or providing a table relating components to functional blocks.

d. Functional diagrams shall show hardware boundaries by using solid, dashed, or dotted lines; various line weights; or different colors or shades (when approved by the contracting activity, see 5.2).

5.7.6 Pictorial diagrams. (See figure 21.)

a. Pictorial diagrams shall show the physical view of components and show relative location and size when doing so aids understanding of relative information. When necessary to portray position or relative location, other equipment items may be shown in phantom.

b. All maintenance significant functional components shall be identified with leader lines.

c. Arrowheads shall be used to show direction of mechanical action or fluid flow.

d. Data shall be presented from the user's viewpoint by only using orientations that are clearly visible to the user.

e. Unless requirement is specifically excluded by the contracting activity, waveform data shall not be shown. When required, waveform data shall be shown pictorially and all necessary data shall be shown. If not provided on functional and schematic diagrams and required in support of maintenance tasks, waveform diagrams shall show the waveforms and nominal values at designated points for

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normally functioning equipment or systems as seen on an oscilloscope.

5.7.7 Cutaway diagrams. Cutaway diagrams (conventional cutaway techniques) shall be used only when necessary to show the internal functioning or flow. All fluid flow shall be identified using patterns or color (when approved by the contracting activity, see 5.2). (See figure 11.)

5.7.8 Logic diagrams. (See figure 22.)

a. Logic diagrams shall show digital circuitry operation. Graphic symbols from ANSI/IEEE STD 91-1984 shall be used. If the logic circuit has no specified symbol, it shall be identified with a rectangle that is labeled to show all circuit functions.

b. Power and clock connections shall be identified in a truth table or connect using breakoff signal.

c. When necessary for clarity, a truth table or timing diagram shall be used. It shall be integrated into the logic diagram or referenced from another diagram. For logic functions, the truth table or timing diagram may be shown inside of a block to describe the relation of input to output signals.

5.7.9 Simplified diagrams. Simplified circuitry and/or simplified functional divisions shall indicate as excluded or included components in the diagram title (for example, "Figure 3. Simplified Transmit Circuit in Standby (Non-Transmit, Non-Receive) Condition (All Relays Unoperated).").

5.7.10 Partial diagrams. Partial diagrams shall show all circuit details completely and reference all destinations of input or output connections (figure 23).

5.7.11 Indicator diagrams. Indicator diagrams shall show all inputs and controls involved in activating the indicator(s).

5.7.12 Test diagrams. Test diagrams shall show test stimuli, item (or circuitry) under test, and test measurement components. When diagrams exist for the item under test, a block diagram representation may be used. In test TMs, the Item Under Test is emphasized (shown in detail); in test equipment maintenance TMs, the test equipment is emphasized.

5.7.13 Power distribution diagrams. Power distribution diagrams shall depict components involved in power input, power form generation, and power distribution. They shall be grouped by power flow. (See figure 24.)

5.7.14 Pyramid diagrams. Pyramid diagrams are a set of interrelated diagrams which shall consist of: a master block diagram, detailed block diagrams, and schematic diagrams. If the

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equipment covered is complex, several levels of detailed block diagrams may be required (figure 25).

5.8 Wiring and interconnection data. TMs supporting systems which have wiring or cabling shall include interconnection information in one or more forms such as cable diagrams, wiring diagrams (DOD-STD-863), or wire lists. All interconnection information shall have the following characteristics:

a. Information keyed to the appropriate maintenance level.

b. User experience level requirements met.

c. Diagrams simple and straightforward.

d. Only required data on diagrams.

5.8.1 Cable diagrams. Cable diagrams shall be provided if the technician must install or remove cables when performing installation, assembly, disassembly, modification, or service. Cable diagrams shall provide all the information necessary to make the electrical connection between assemblies, chassis, bays, units, and systems in an easily understood format. Each cable diagram shall consist of an illustration and accompanying table. If cable routing is of a special nature, it shall be so noted. For very complex systems, where routing is of great importance, additional diagrams showing desired cable locations may be necessary. (See figure 26.)

5.8.1.1 Cable diagram illustration. The illustration shall show all related connectors. Assembly names and jack numbers shall be listed (figure 26).

5.8.1.2 Cable diagram table. (See figure 26.) The table shall meet the following requirements:

a. Cable entries shall be listed in numerical order or by preferred connection sequence.

b. Cable origin shall precede cable destination.

c. Cable origin and destination shall include assembly name, assembly jack number, and cable plug number.

d. Both list and illustration shall appear on the same page or facing pages.

e. For simple equipment, a table may not be needed and a pictorial diagram that actually shows the routing may be substituted.

5.8.2 Wiring diagram. Wiring diagrams shall be provided for medium complex assemblies in which the technician must fault isolate and repair wiring. If complete coverage of wiring is included in the schematic, no special wiring diagrams shall be required. Wiring diagrams shall provide an illustration of

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signal flow. They may also be used to show how to connect wires, when required. (See figure 27. See also ANSI Y14.15.) Usable wiring diagrams shall include the following requirements:

a. Limit each drawing to an individual system to eliminate overly large foldout illustrations that would be confusing to the user.

b. Show point-to-point wiring between connectors and terminals.

c. Show continuity of wiring through junction boxes and other equipment to permit following the wires from their origin to their termination.

d. Use terminal, connector, and wire identifiers that appear on the hardware.

e. Assign signal codes to wires to help the user follow signals in equipment that has complex wiring.

f. Show only the wiring that is appropriate for the level of maintenance for which the TM is written.

g. Include a list of components giving the reference designation, nomenclature, location, and access cover (as applicable) for every component on each diagram.

h. Use heavier lines for component outlines than for wires.

i. Label every wire or wire segment with its wire number.

j. Minimize crossing lines with a layout that has no more than 20 line intersections in any 4-inch square area.

k. Avoid turning any wiring diagram sideways (90 degrees) on a page.

l. Use a logical, easy-to-follow technique to show the wire continuation onto other diagrams.

5.8.3 Wire list. Unless requirement is specifically excluded by the contracting activity, wire lists shall be provided for cables and complex wire runs that the technician must fault isolate and repair. Connector keying diagrams shall be included when necessary. Wire lists shall provide wiring data in tabular format. (See figure 28. See also ANSI Y14.15.) Usable wire lists shall include the following requirements:

a. Create one table that is in hardware number order.

b. Create another table that is in signal code number order.

c. List from wire origin to wire destination.

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d. List terminal, connector, and wire identifiers that appear on the hardware.

e. List only the wiring that is appropriate for the level of maintenance for which the TM is written.

5.9 Illustrations for procedures. An illustration for operator's or maintenance procedures shall have one or more text steps with each illustration. It is not necessary to illustrate each step of a maintenance procedure, such as the removal of screws with an ordinary screwdriver, lifting off a cover after the screws have been removed, etc. Illustrations for procedures should supplement the text by clarifying procedures which are of a special nature or are not obvious. The illustration(s) shall be located as close to the text step(s) as possible. (See figure 5.)

5.9.1 Tool and test equipment illustrations. Standard tools and test equipment shall not be illustrated, nor shall self-evident or generally known uses be shown. Only uncommon or unusual uses and connections for test purposes shall be illustrated if it is essential to do so to avoid misunderstanding. Unusual operations shall be illustrated.

5.9.2 Illustrations of special tools and test equipment. Unless requirement is specifically excluded by the contracting activity, special tools and test equipment shall be illustrated.

5.9.3 Exploded views. Exploded views of the equipment shall be used in RPSTLs and, when practical, to support maintenance instructions. Index numbers, keyed to a legend, list, or text reference, shall be used to identify parts. If the equipment is of such a nature that it cannot be adequately illustrated by a single exploded view, it may be exploded by subassemblies as separate views. In such cases, locator views (See 5.9.4.) shall be used, as needed, to orient the user to the proper area. Items shall be numbered sequentially (in clockwise, disassembly, or text mention sequence, depending upon how the exploded view is used). (See figure 6.)

5.9.4 Locator views. When required by the complexity of the equipment or to assist in user orientation of part(s), detail-view illustrations shall contain a locator view. The view shall consist of a view of the overall equipment or item with the area covered by the view highlighted. (See figure 7.)

5.9.5 Detail views. A detail view of a part or subassembly shall be illustrated when the subject matter cannot be clearly illustrated in the main view. The desired subject matter may be identified with detail letter(s) or index number(s) on the main view and illustrated, as required, in the detail. (See figure 1.)

5.10 Quality assurance provisions. Quality assurance and quality control for TMs, revisions, supplements, and changes shall be in accordance with MIL-STD-361-4.

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5.11 Packaging and preparation for delivery. TMs, revisions, supplements, and changes shall be packaged in accordance with MIL-STD-361-5.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This document prescribes the graphics requirements applicable to various types of technical manuals, revisions, supplements, and changes prepared by or for the Department of the Army.

6.2 Issue of DODISS. When this document is used in acquisition, the applicable issue of the DODISS must be cited in the solicitation. (See 2.1.1.)

6.3 Tailoring guidance. The contracting activity should tailor any required options offered herein in accordance with MIL-M-63008.

6.4 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this standard.

b. Issue of DODISS to be cited in the solicitation, and, if required, the specific issue of individual documents referenced. (See 2.1.1.)

c. Color (other than black and shades of black) allowed (5.2).

d. Type of clothing for illustrated human figures (samples to be provided) (5.3.2).

6.5 Supersession data. This document consolidates, standardizes, and supersedes the graphics requirements contained in the following documents:

MIL-M-49502(TM)	-	Manuals, Technical: Repair Parts and Special Tools List.
MIL-M-49504(TM)	-	Manuals, Technical: Combat Vehicle Precombat Checklist.
MIL-M-63004C(TM)	-	Manuals, Technical: Preparation of Lubrication Orders.
MIL-M-63036D(TM)	-	Manuals, Technical: Operators, Preparation of.
MIL-M-63038C(TM)	-	Manuals, Technical: Organizational and Aviation Unit, Direct Support or Aviation Intermediate, and General Support Maintenance.

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MIL-M-63041D(TM) - Manuals, Technical: Preparation of Depot Maintenance Work Requirements.

6.6 Subject term (key word) listing. The following terms are to be used in identifying this document during retrieval searches:

Callouts
Color in illustrations
Combination illustrations
Continuous tone
Credit lines
Crop and size marks
Exploded view
Final reproducible copy
General illustration requirements
Graphic presentations
Halftones
Leader lines
Legends
Phantom
Pictorial
Steps
Symbols

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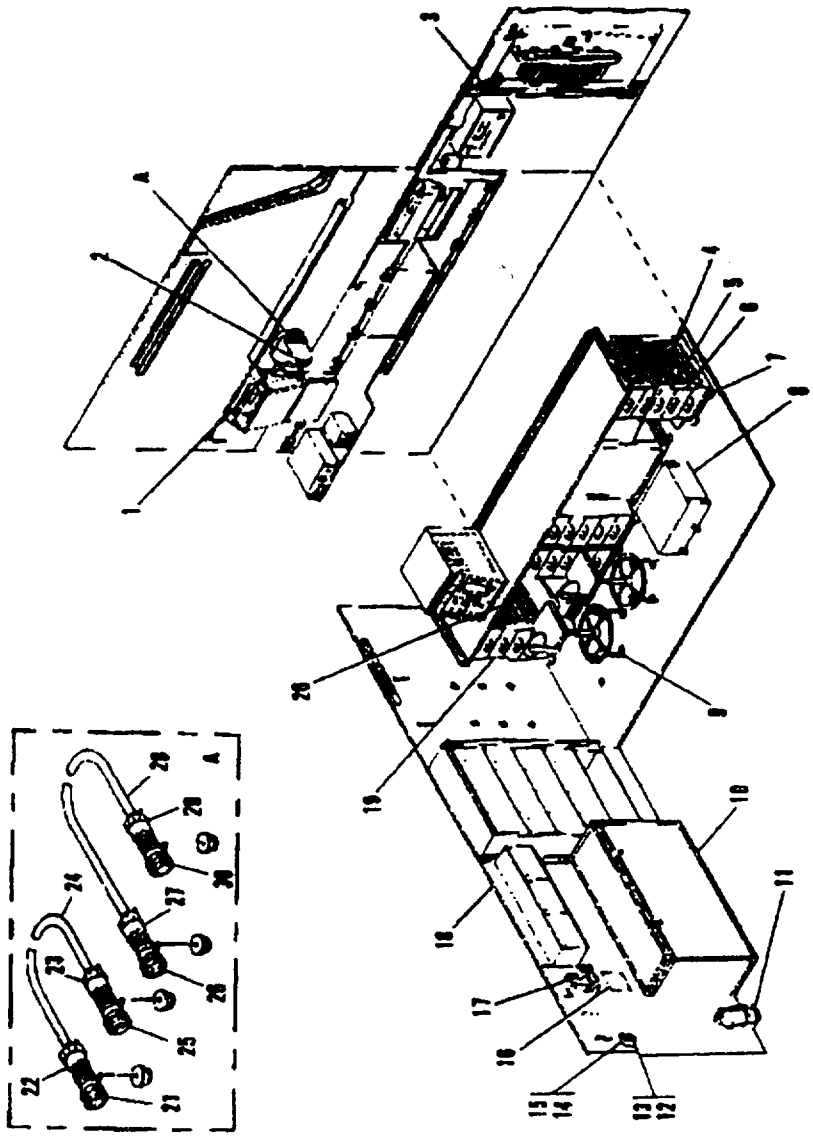
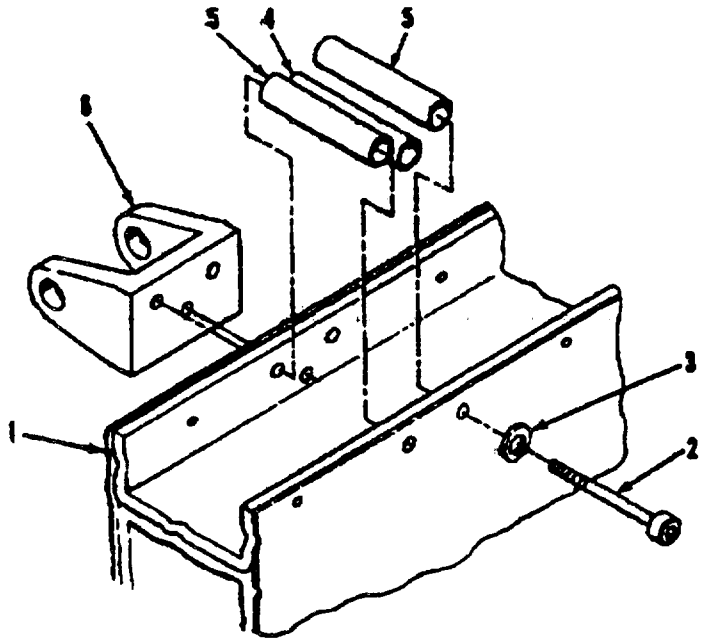


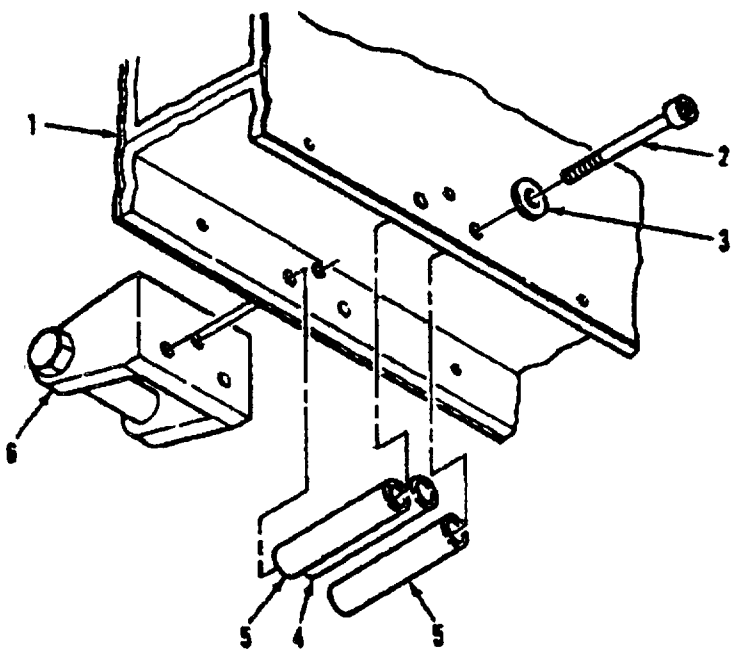
Figure 1-1. Interior of Shelter Major Components.

FIGURE 1. Example of illustration turned sideways on page (including a detail view) (reference 4.1.1b, 5.5.4c, 5.9.5).

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A. UPPER MOUNT



B. LOWER MOUNT

FIGURE 2. Example of multisection illustration with border rules (reference 5.1.6a, 5.3.1).

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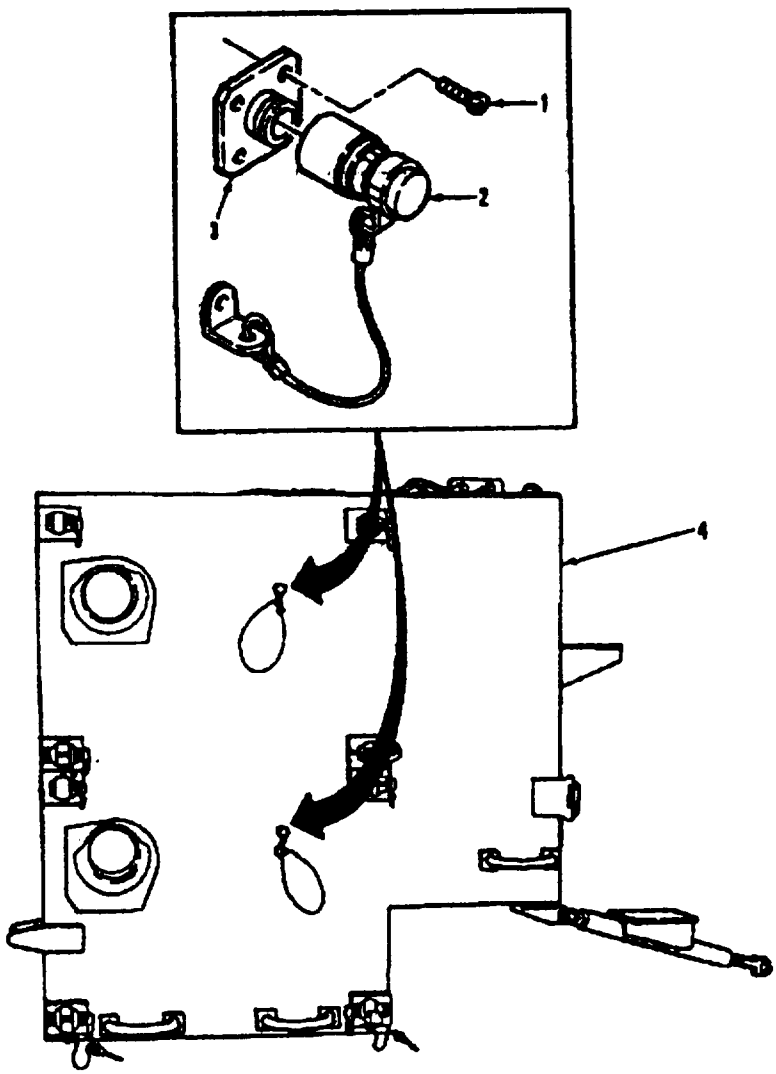


FIGURE 3. Example of boxes, exploded view, and locator view (reference 5.3.1, 5.3.4.1a).

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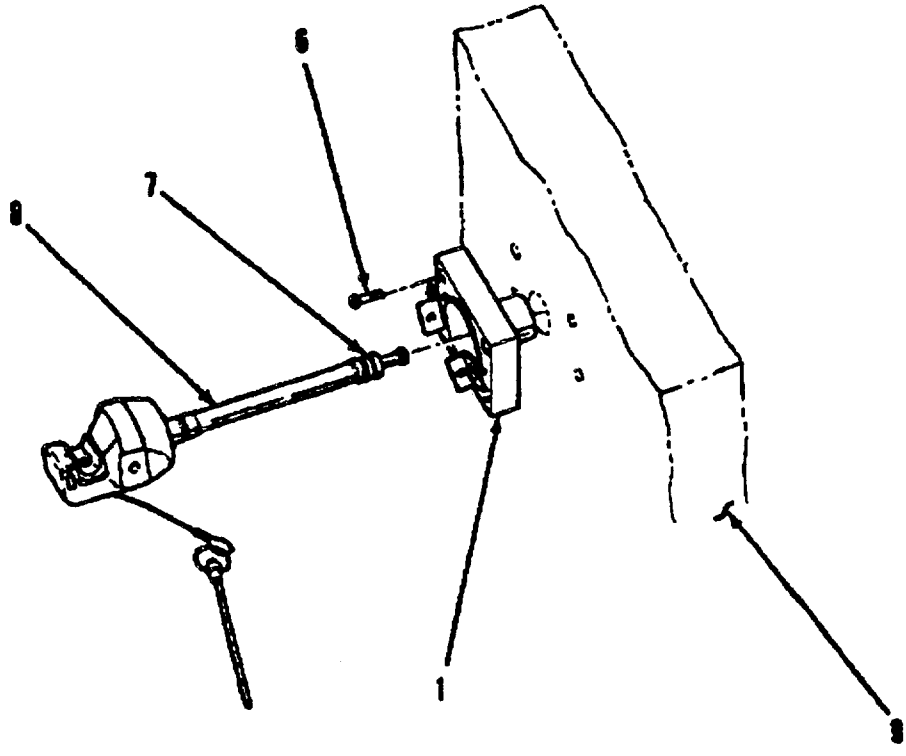
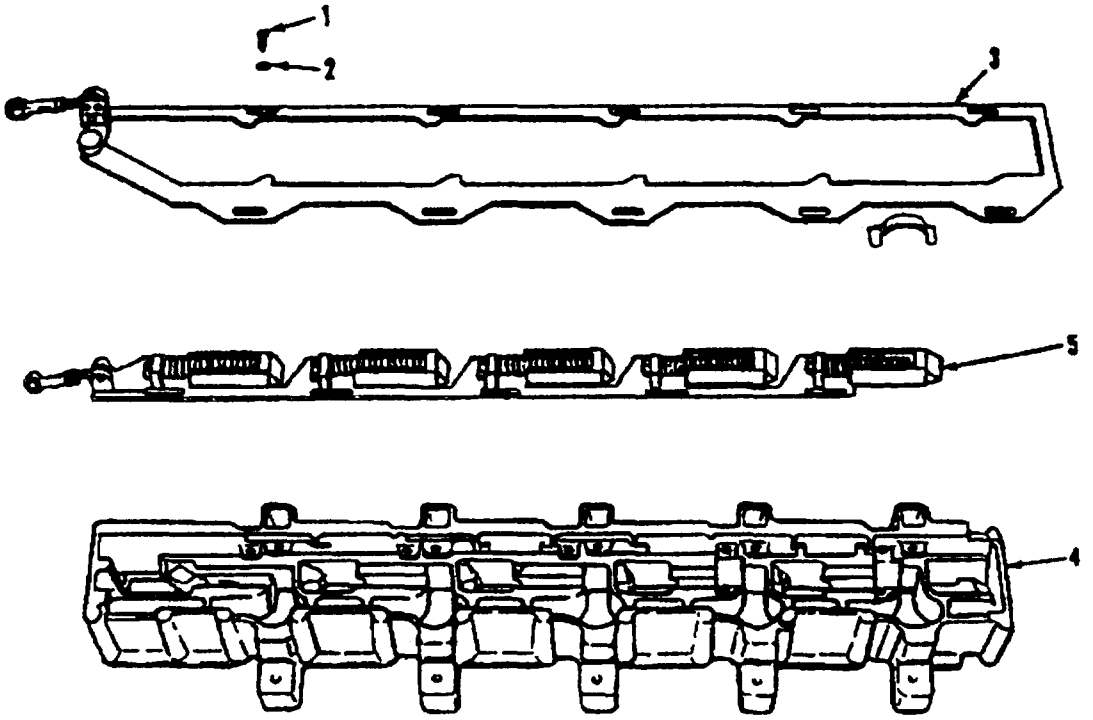


FIGURE 4. Example of items in phantom (reference 5.3.4.1a).

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3.42 LATCHING AND ARMING LEVER REPLACEMENT. - Continued



3.42.1 Removal.

a. Remove latching lever per paragraph 2.44.

b. Remove arming lever per paragraph 2.45.

c. Remove wiring harness cover per paragraph 3.38.

d. Remove eight shoulder screws (1) and washers (2) attaching arming slide (3) to housing (4).

e. Lift arming slide (3) and latching slide (5) from housing (4).

3.42.2 Installation.

a. Place arming slide (3) and latching slide (5) in housing (4).

b. Install eight shoulder screws (1) and washers (2) through arming slide (3) and latching slide slots into housing (4).

c. Install wiring harness cover per paragraph 3.38.

d. Install arming lever per paragraph 2.45.

e. Install latching lever per paragraph 2.44.

FIGURE 5. Example of illustrations with procedures (reference 5.3.4.1b, 5.5.4a, 5.5.4b, 5.9).

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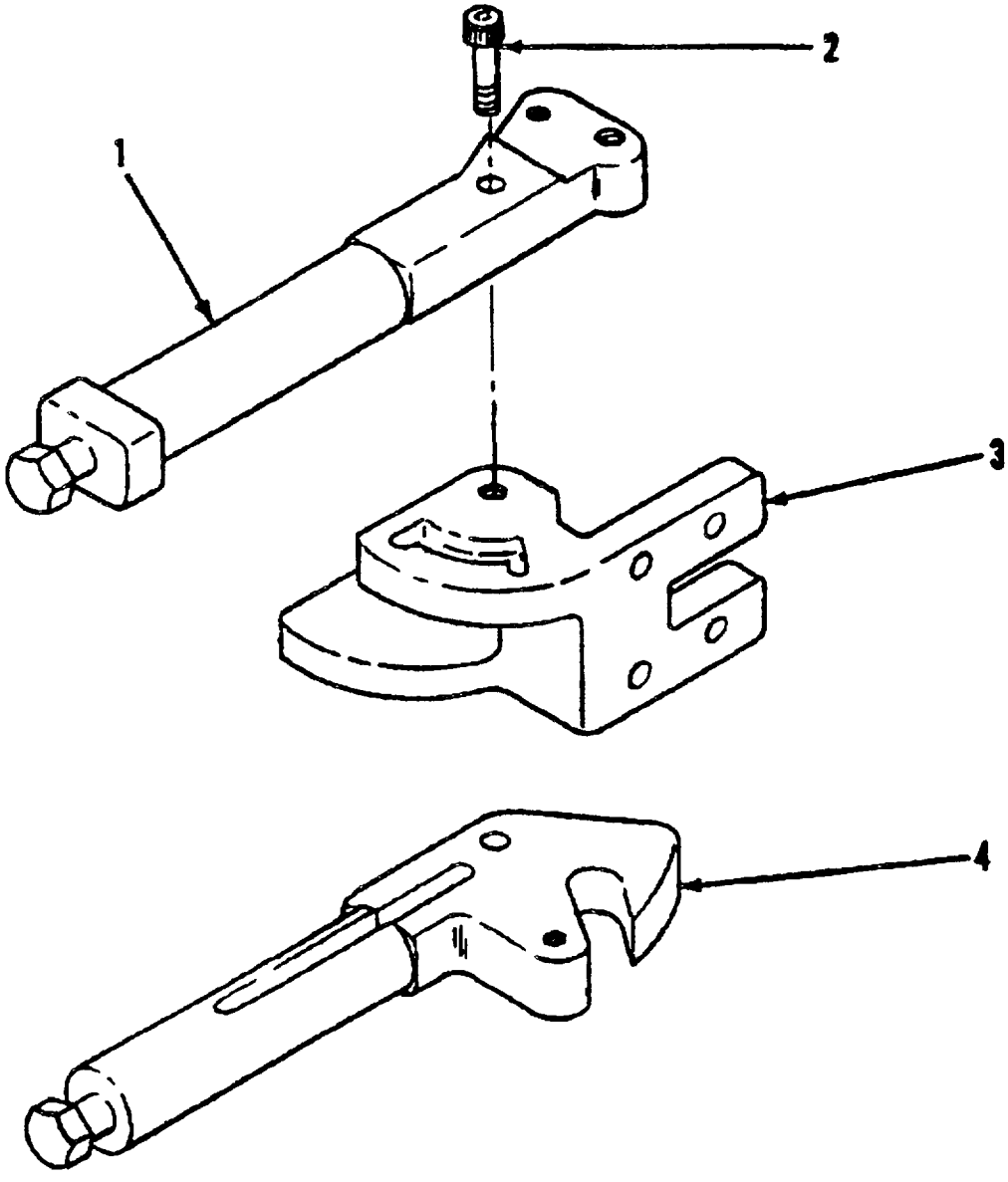


FIGURE 6. Example of exploded items with clockwise numbering (reference 5.3.4.1c, 5.9.3).

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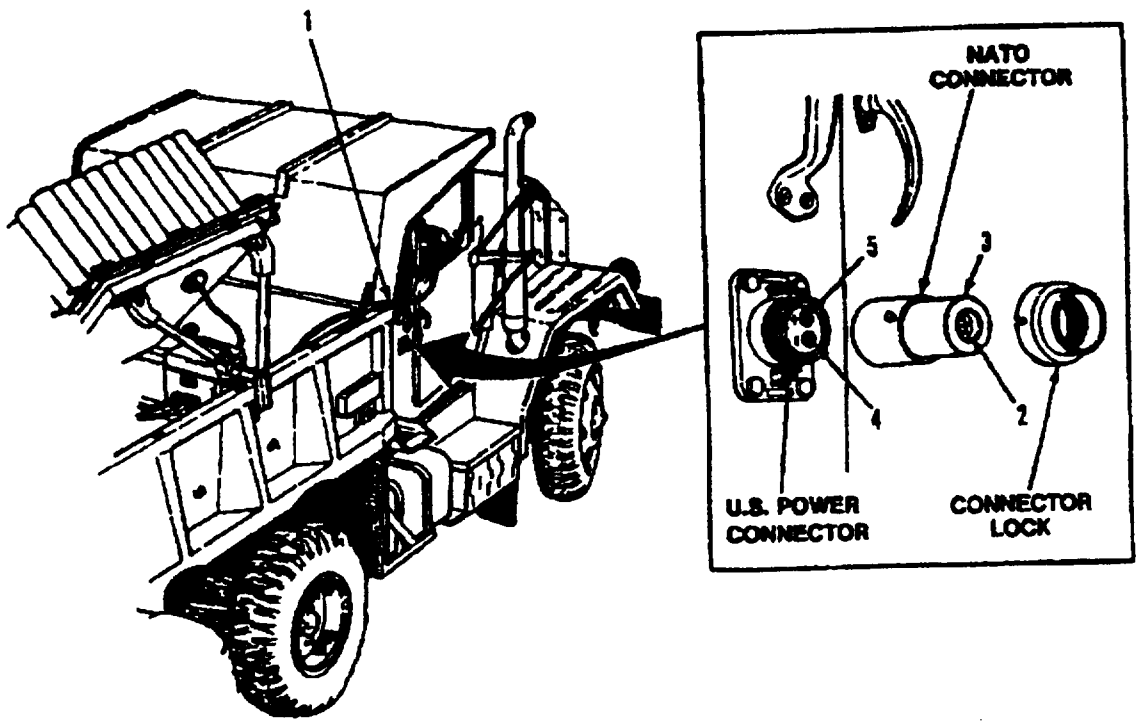


FIGURE 7. Example of locator illustration with nomenclature, leader lines, and sweep arrows (reference 5.3.4.2, 5.3.4.3, 5.5.4, 5.9.4).

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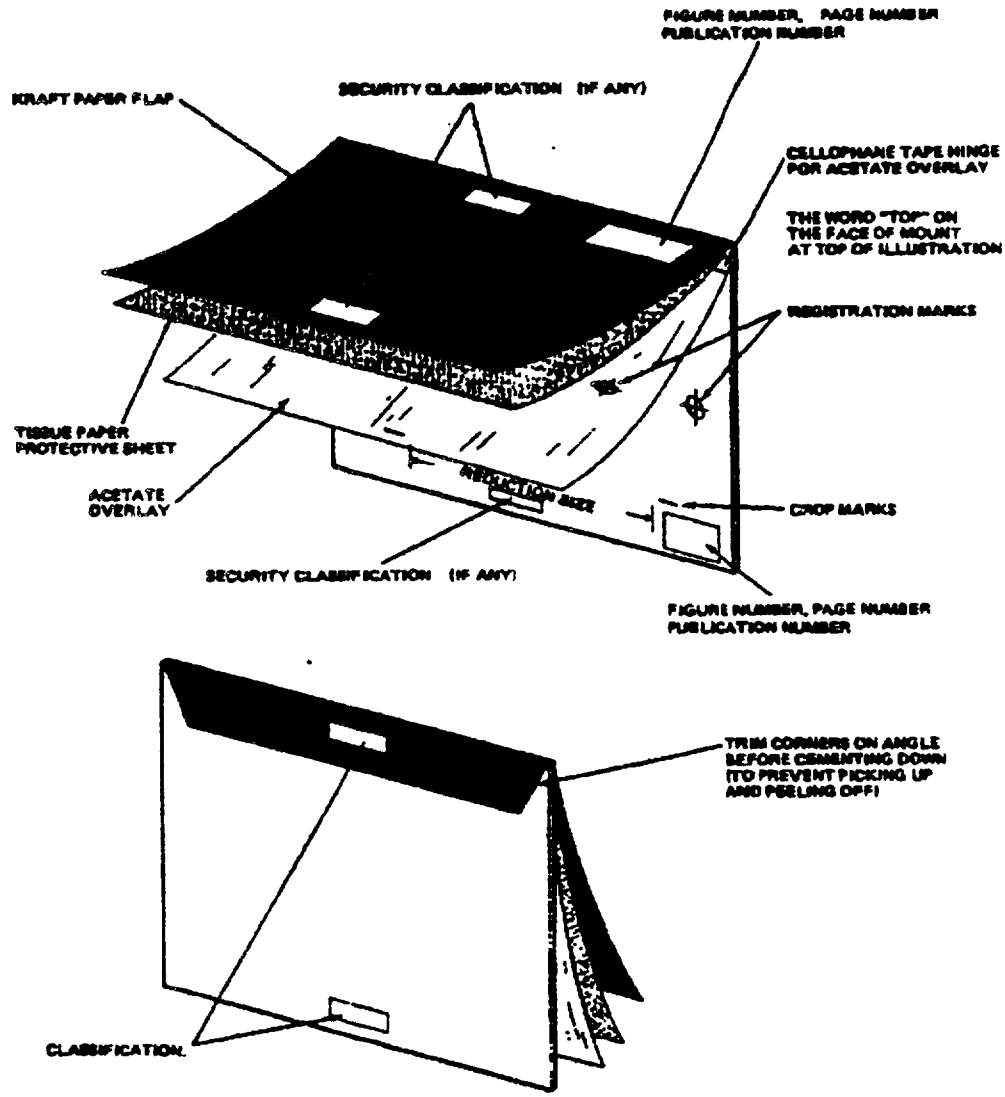


FIGURE 8. Identification, marking, and protective covering for artboard (reference 5.3.5, 5.4.2, 5.4.3, 5.4.4, 5.4.5).

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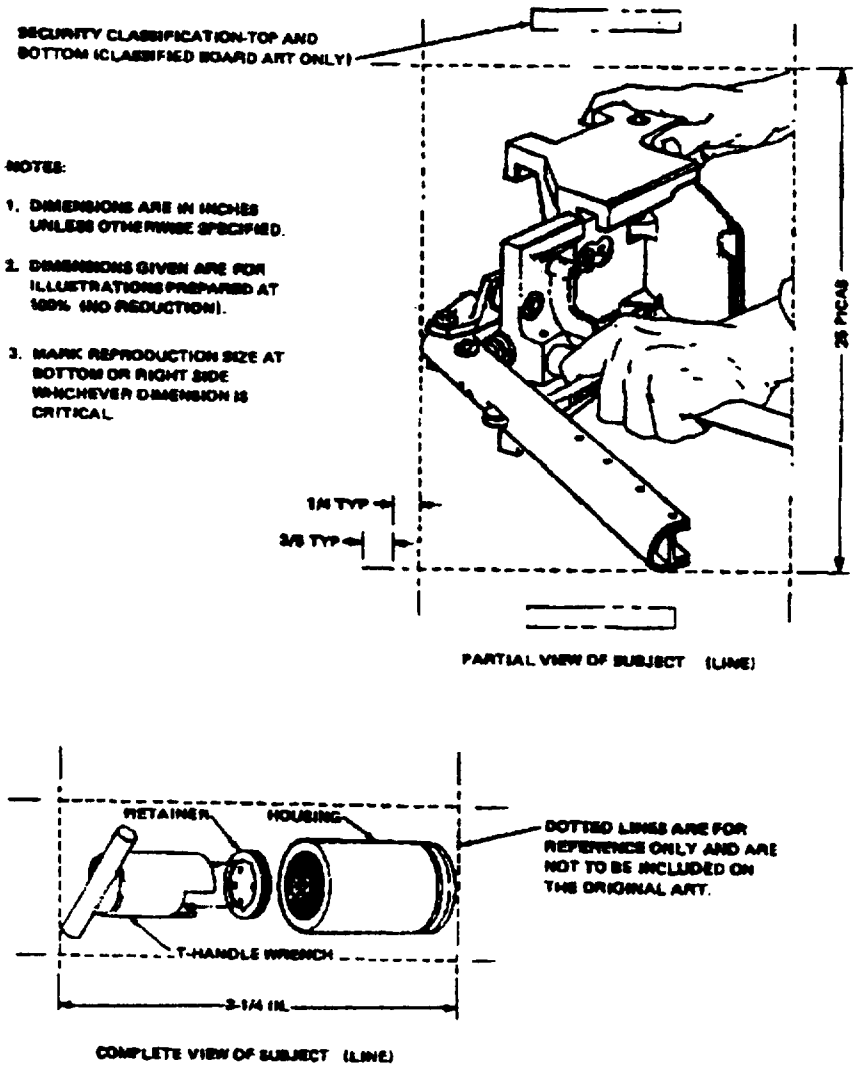


FIGURE 9. Cropping and sizing of illustrations (reference 5.4.1).

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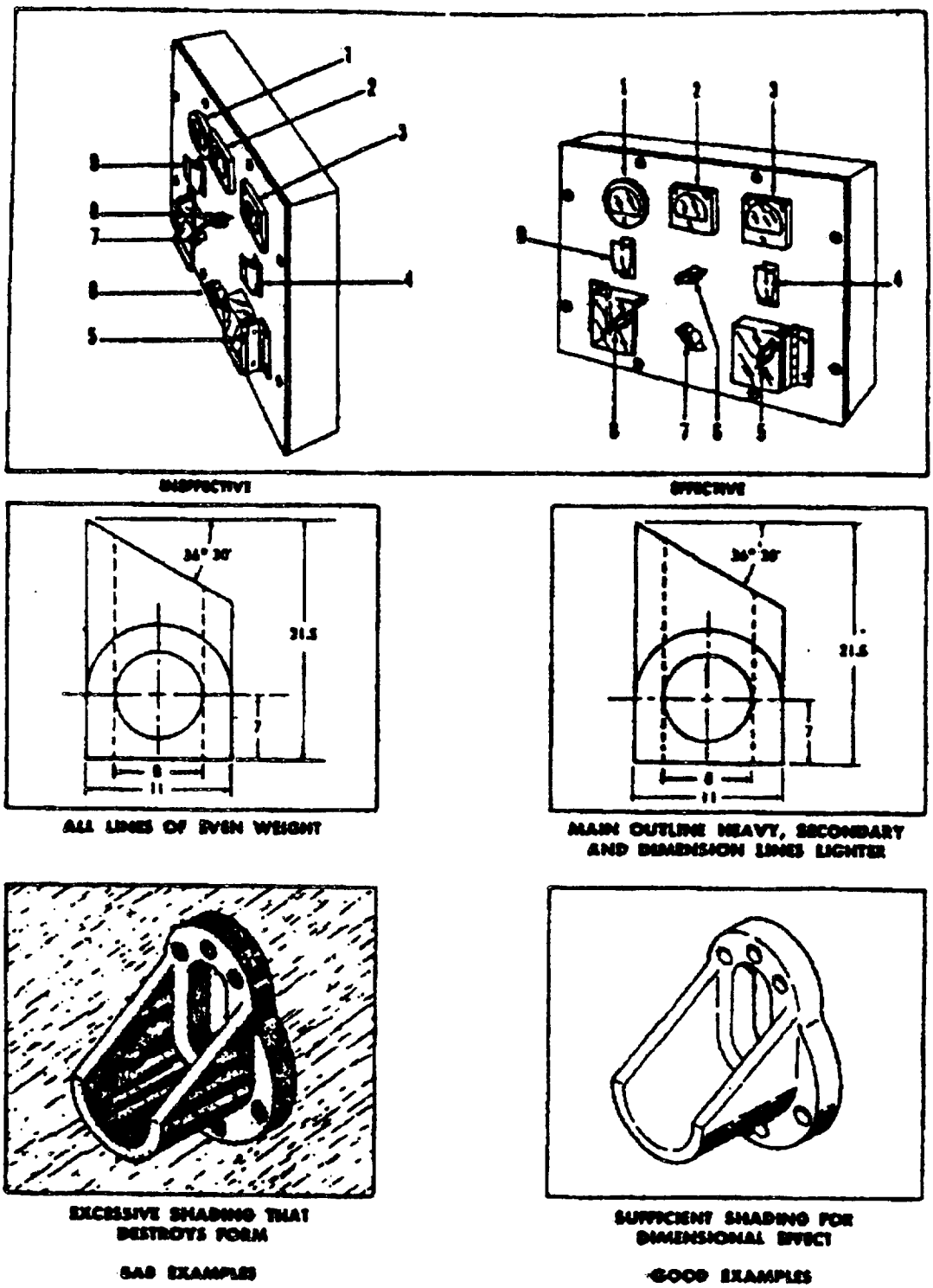


FIGURE 10. Examples of good and bad angle view, line weight, and shading on line drawings (reference 5.5.3).

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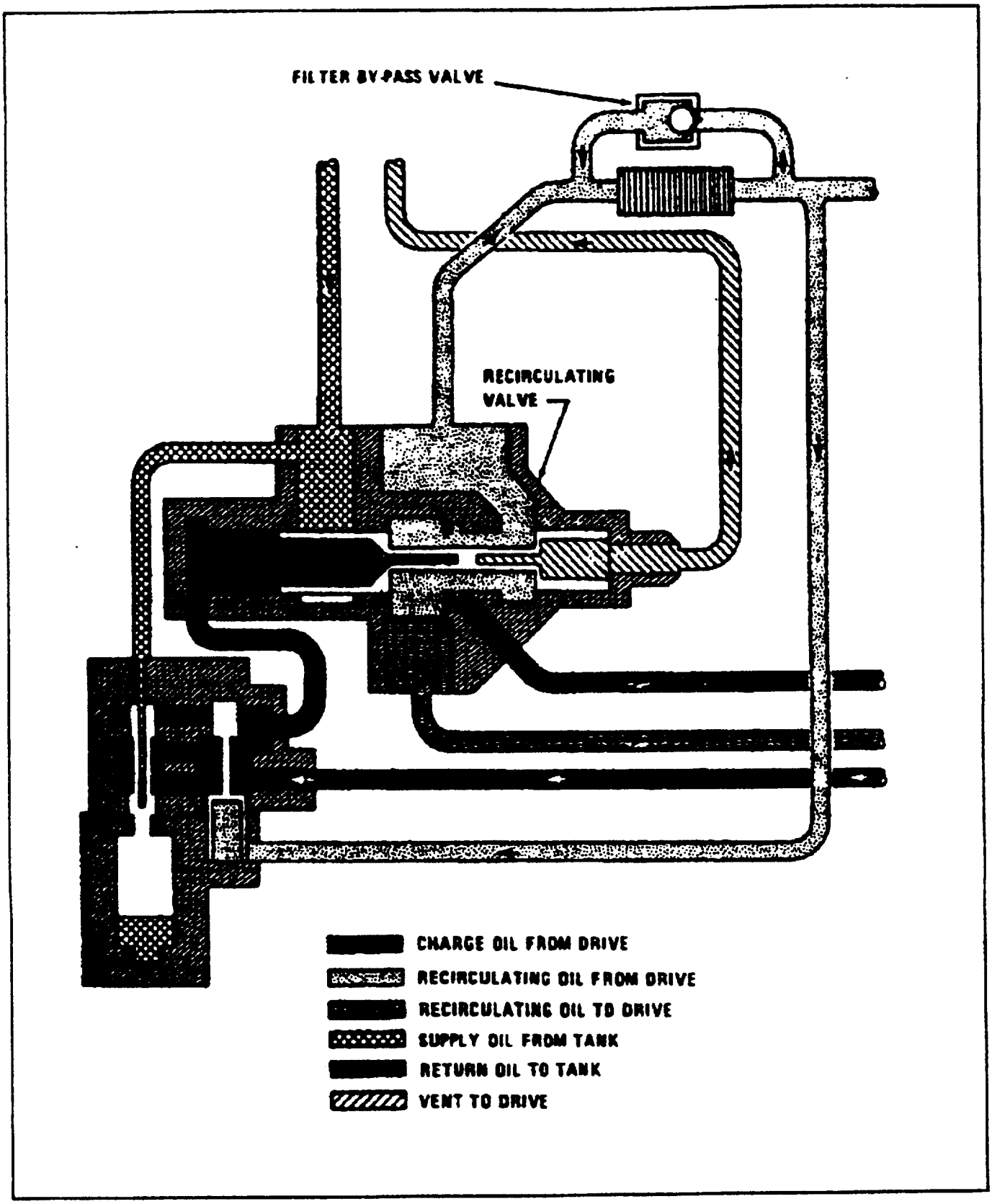


FIGURE 11. Example of use of patterns instead of colors and cutaway drawing showing fluid flow (reference 5.5.3, 5.7.7).

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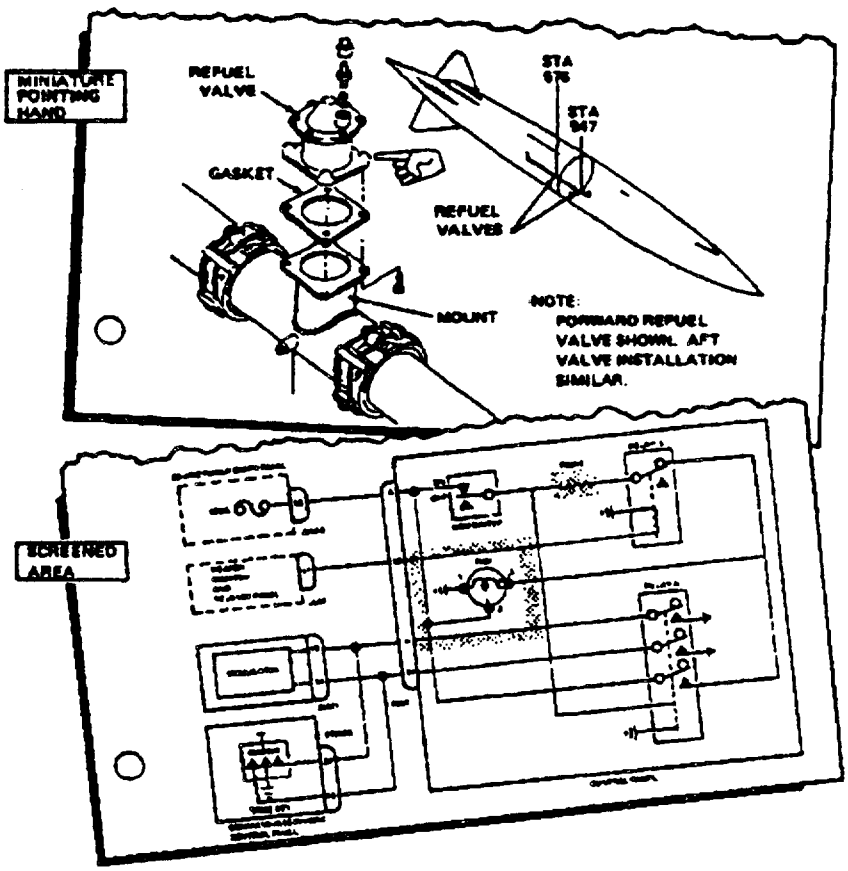
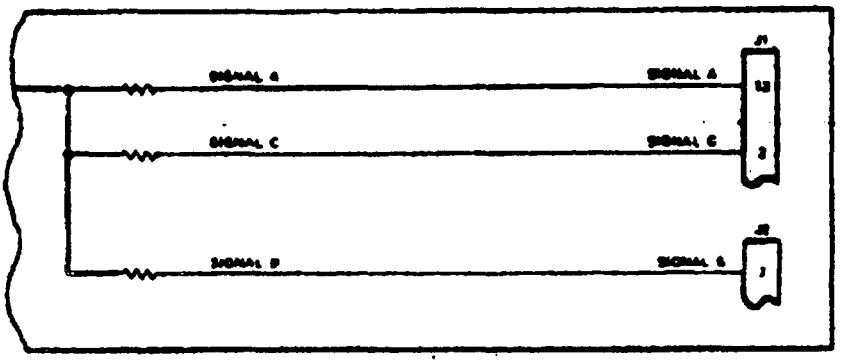
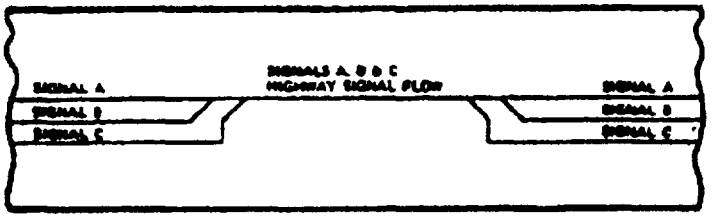


FIGURE 12. Example of change symbols (reference 5.6.2).

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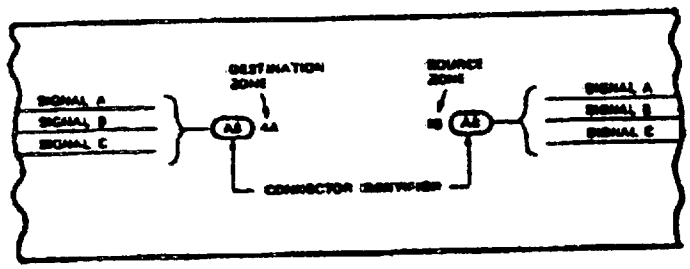
POINT-TO-POINT METHOD FOR SIGNAL FLOW



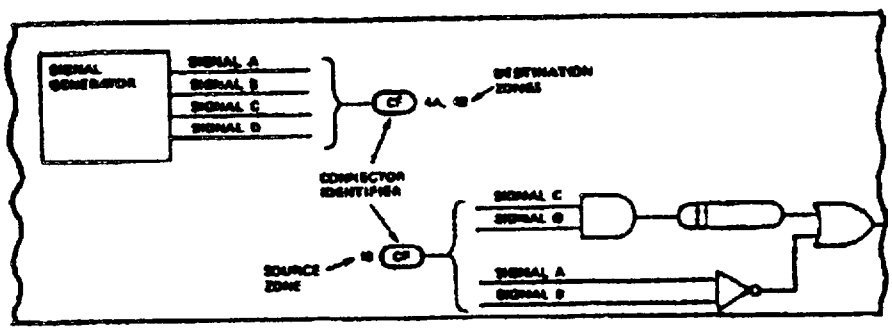
HIGHWAY METHOD FOR SIGNAL FLOW

FIGURE 13. Example of highway and point-to-point methods for signal flow (reference 5.7.4.1a, 5.7.4.1b).

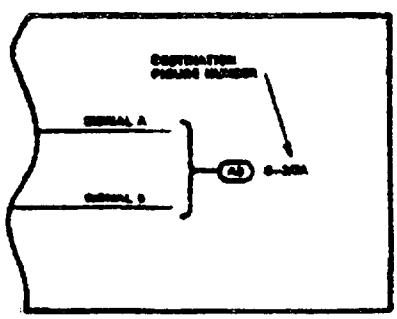
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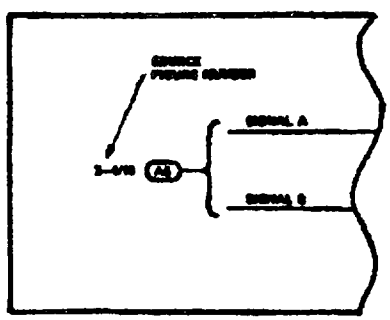
OVAL CONNECTORS USED TO CONTINUE SIGNALS



OVAL CONNECTORS USED TO CONTINUE SIGNALS TO MORE THAN ONE DESTINATION



(Figure 2-4)

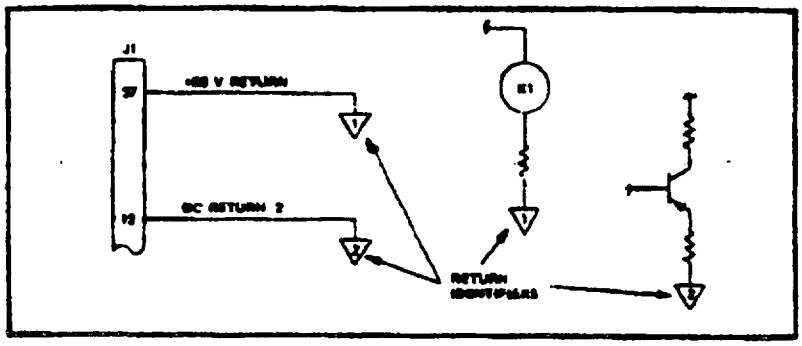


(Figure 8-3)

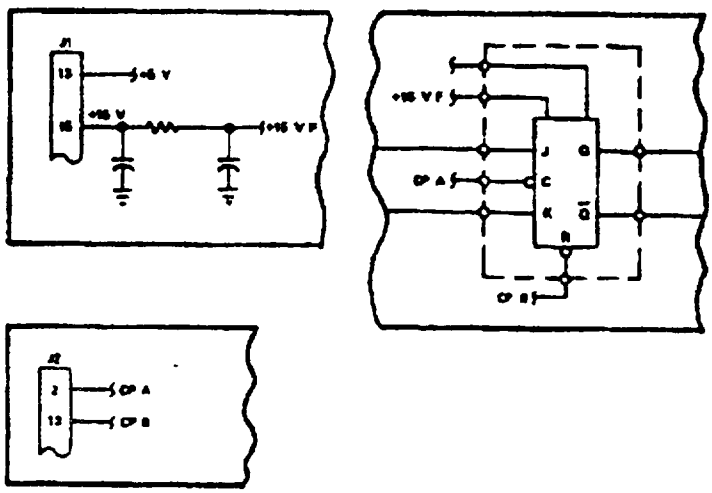
OVAL CONNECTORS USED TO CONTINUE SIGNALS BETWEEN DIAGRAMS

FIGURE 14. Example of oval connectors to continue signals - single sheet and multisheet diagrams (reference 5.7.4.1.1a, 5.7.4.1.2b, 5.7.4.1.3b).

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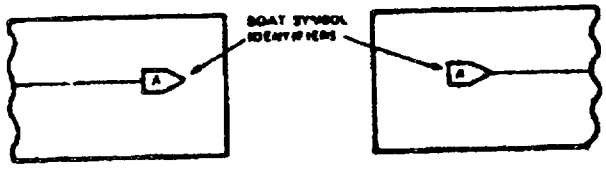
SIGNAL RETURNS USED TO CONTINUE SIGNAL



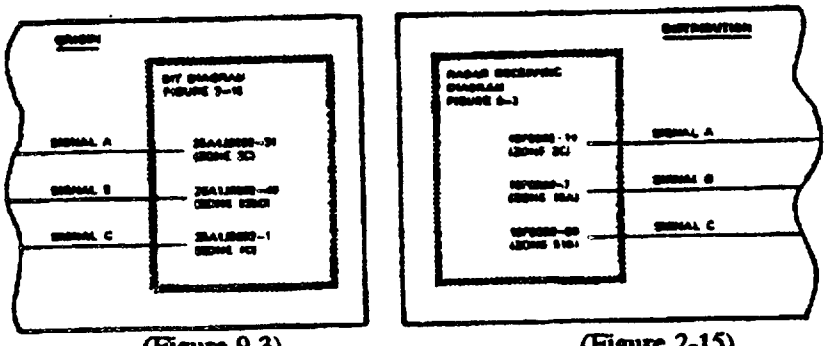
BREAKOFF SYMBOLS USED TO CONTINUE SIGNALS

FIGURE 15. Example of signal returns and breakoff symbols to continue signals - single sheet diagram (reference 5.7.4.1.1b, 5.7.4.1.1c).

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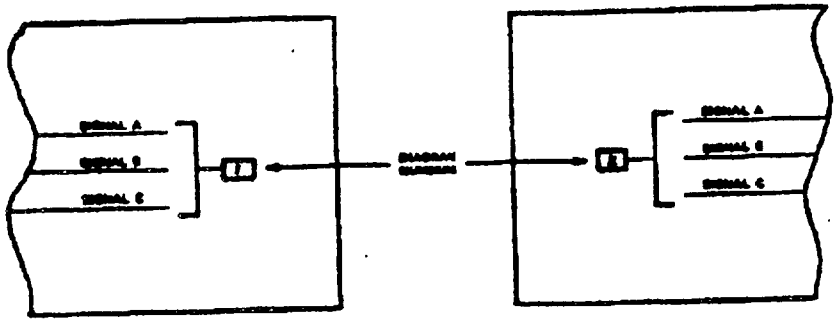


BOAT SYMBOLS USED TO CONTINUE SIGNALS
BETWEEN ADJACENT SHEETS OF A DIAGRAM



(Figure 9-3) (Figure 2-15)

BLOCK TECHNIQUE FOR CONTINUING SIGNALS BETWEEN DIAGRAMS

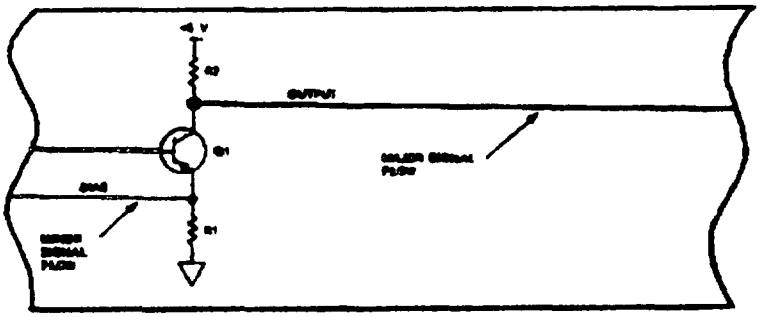


(Diagram 2) (Diagram 1)

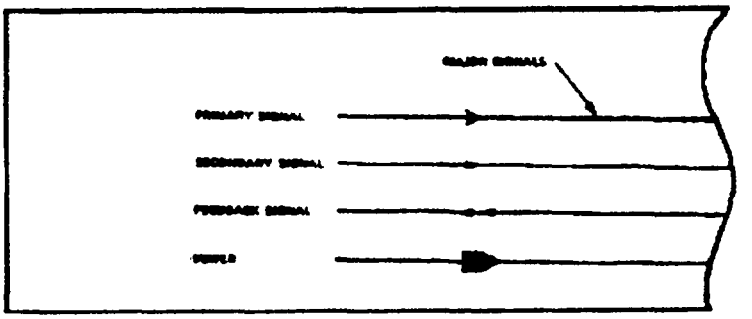
PYRAMID DIAGRAMS INTERDIAGRAM SIGNAL CONTINUATION

FIGURE 16. Example of boat symbols, block technique, pyramid method to continue signals - multisheet diagram (reference 5.7.4.1.2a, 5.7.4.1.3a&c).

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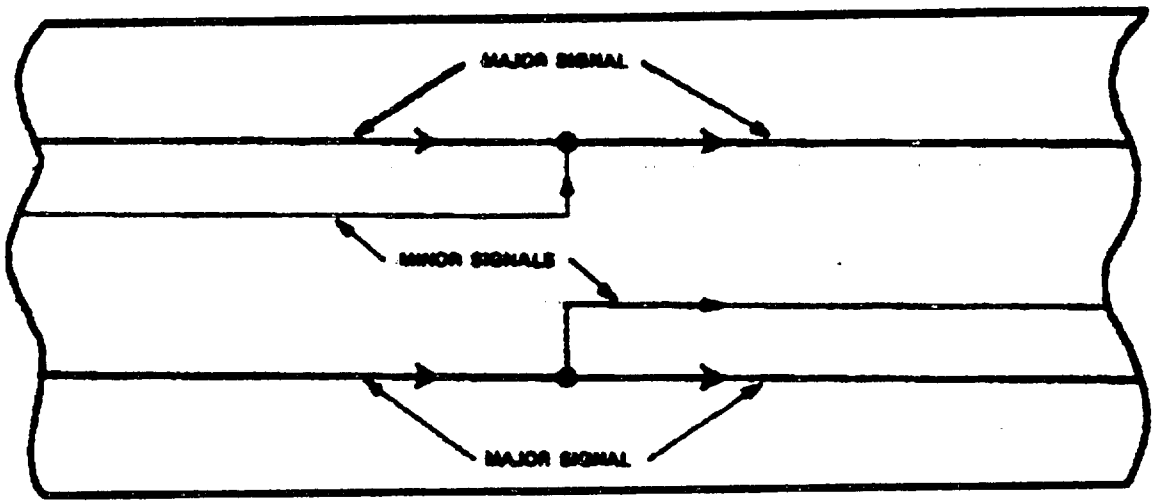
LINE WIDTH CODE USED TO INDICATE SIGNAL IMPORTANCE



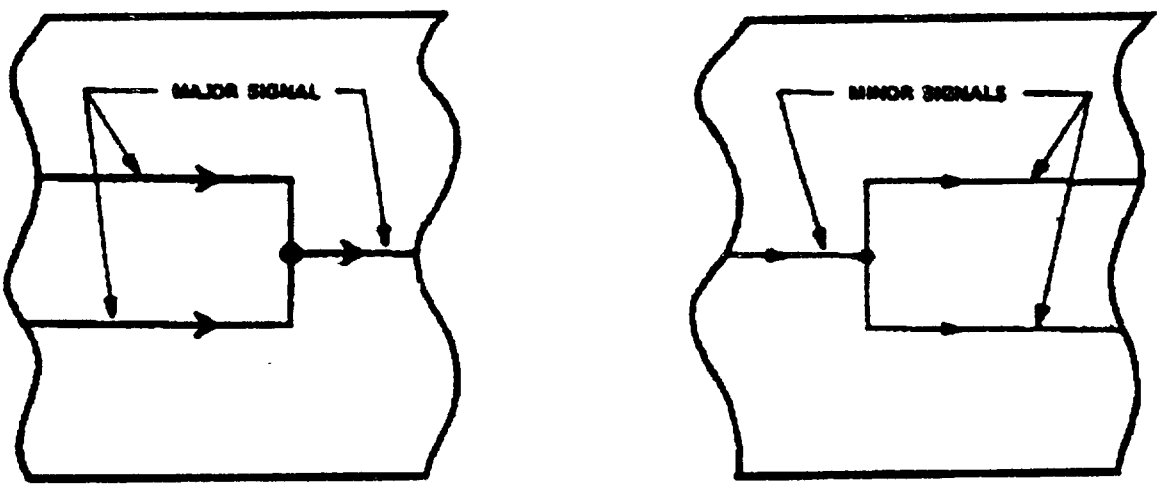
ARROWHEADS USED TO INDICATE SIGNAL TYPES

FIGURE 17. Example of signal difference techniques (reference 5.7.4.2).

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SUBORDINATE JUNCTIONS USED TO INDICATE DIFFERENCES IN SIGNAL IMPORTANCE



COORDINATE JUNCTIONS USED TO INDICATE EQUALITY IN SIGNAL IMPORTANCE

FIGURE 18. Example of subordinate and coordinate junctions (reference 5.7.4.3).

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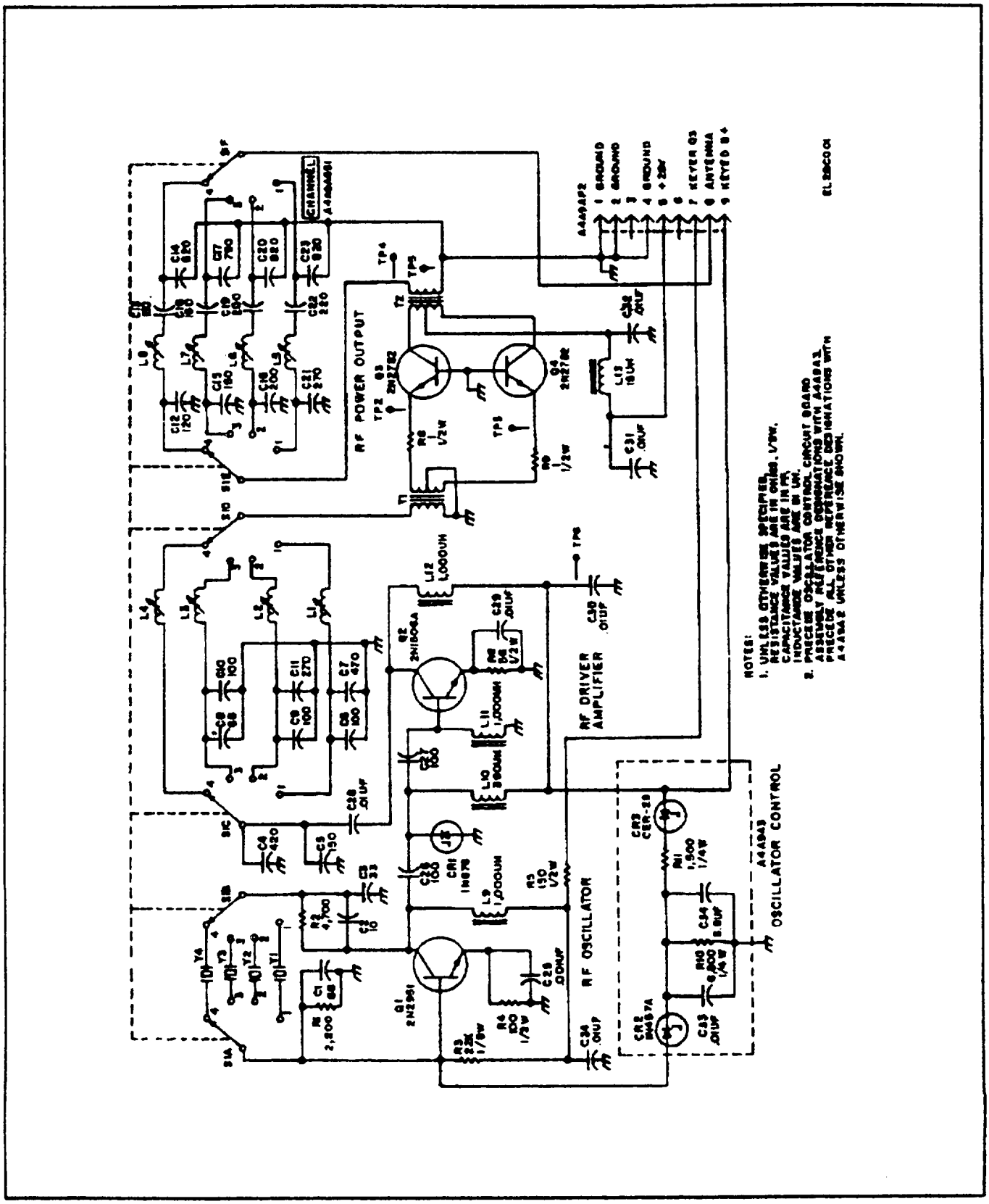


FIGURE 19. Example of schematic diagram (reference 5.7.5.1).

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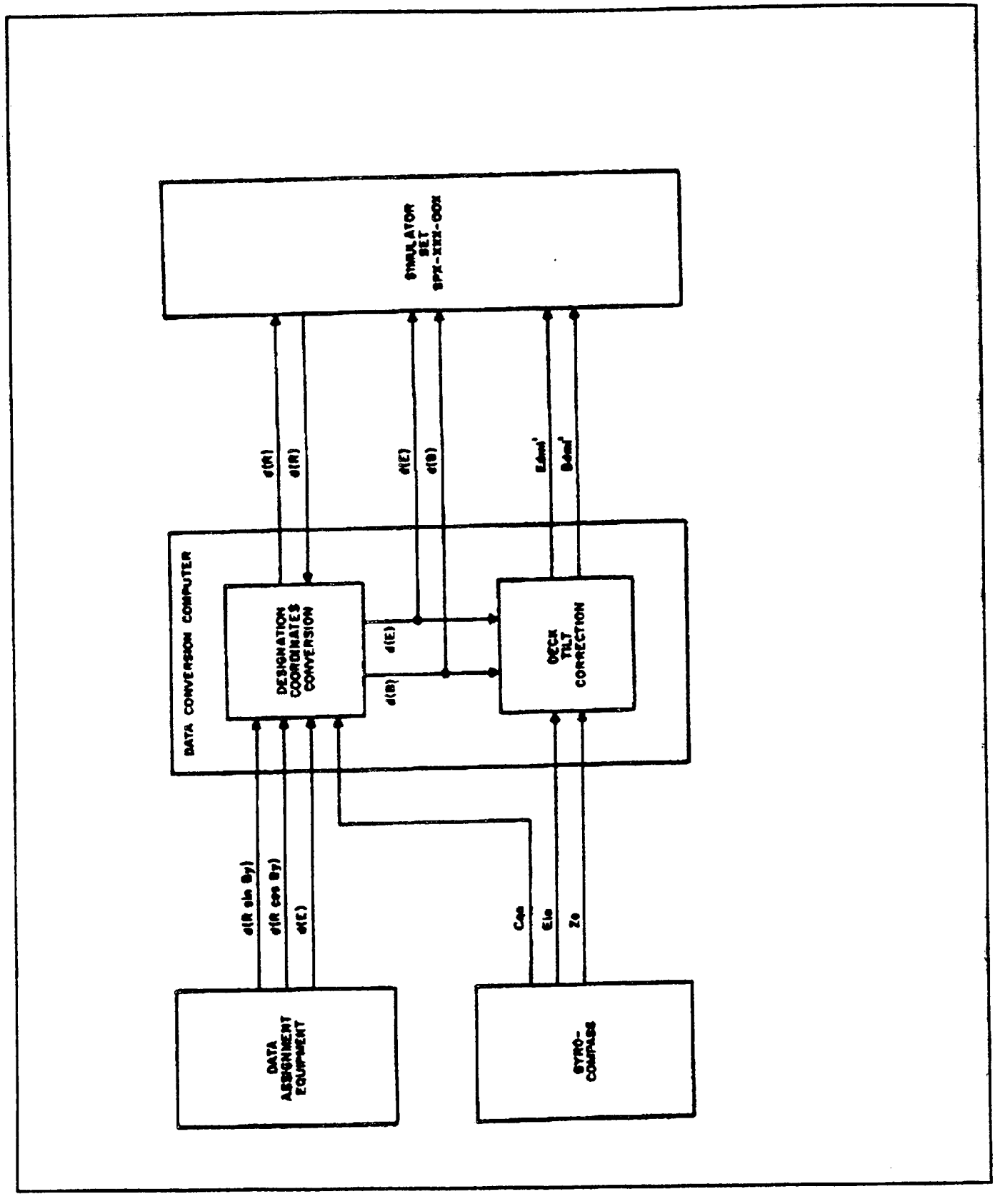


FIGURE 20. Example of functional diagram (reference 5.7.5.2).

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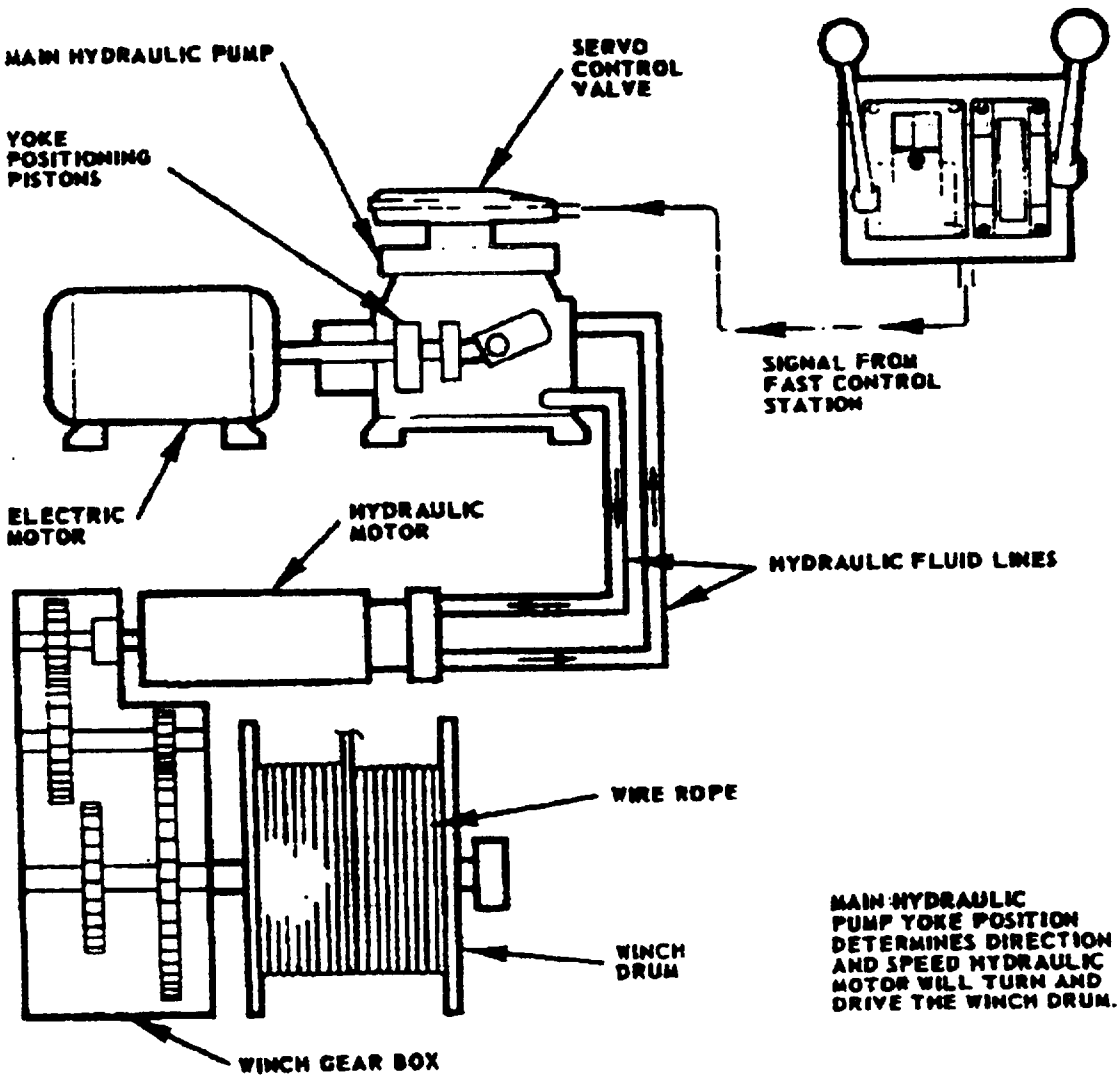


FIGURE 21. Example of pictorial diagram (reference 5.7.6).

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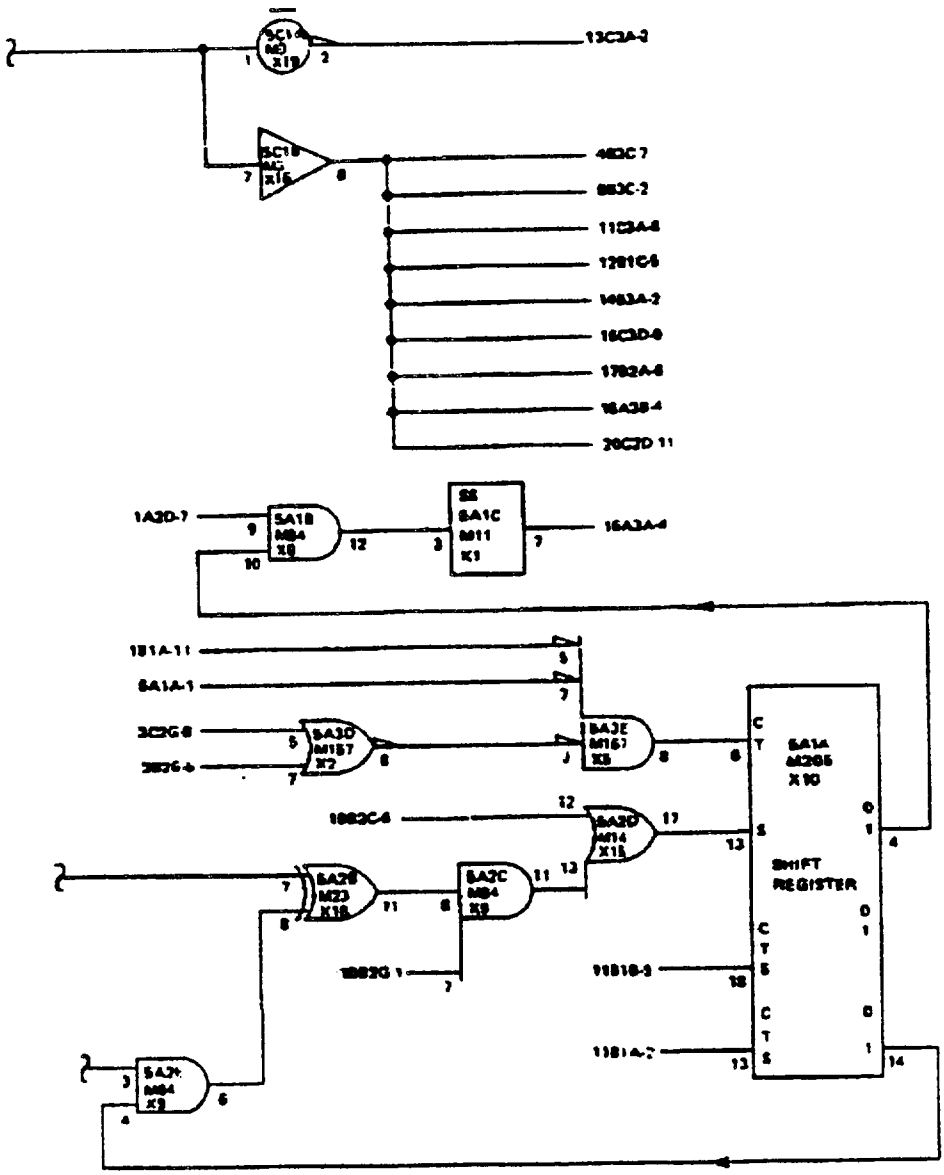


FIGURE 22. Example of logic diagram (reference 5.7.8).

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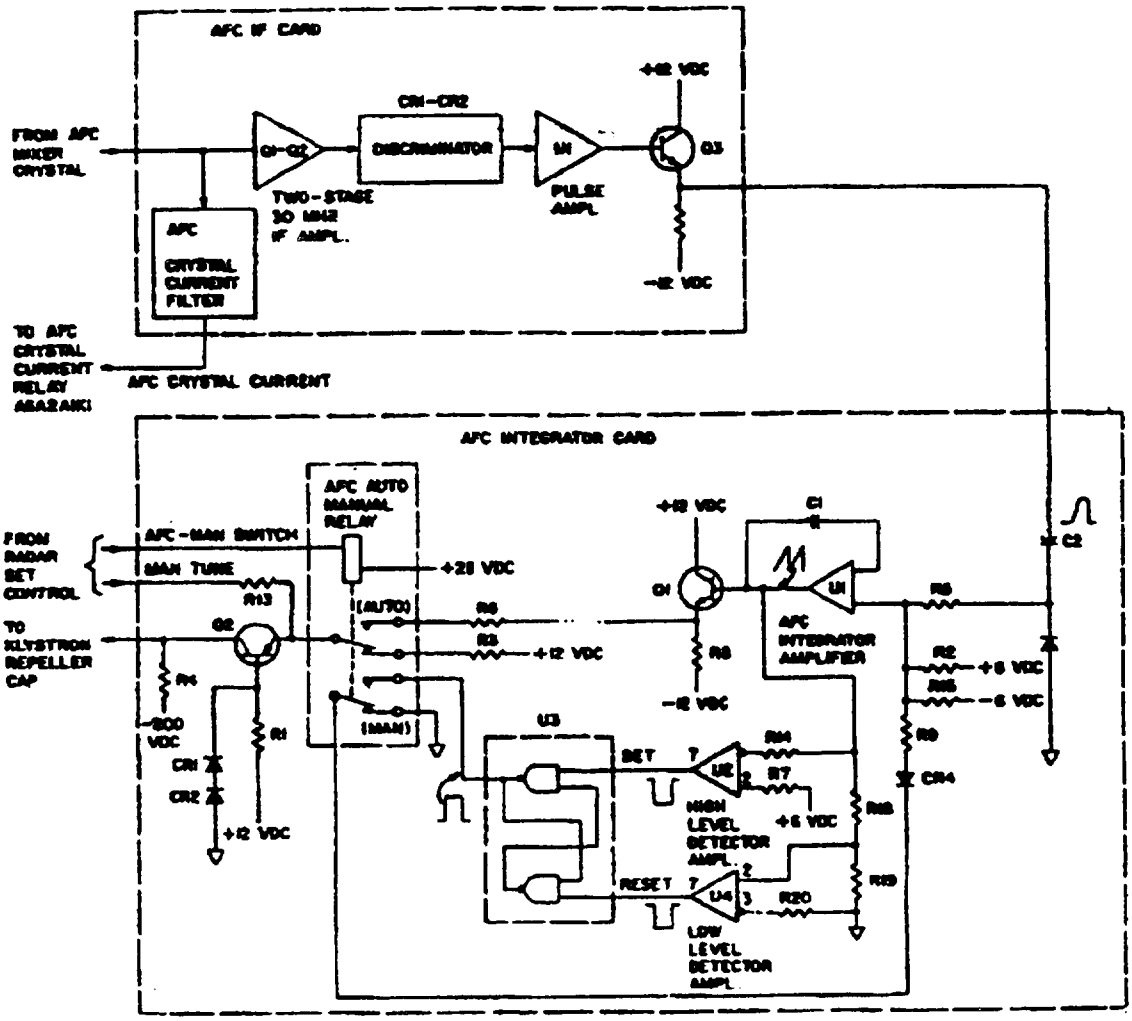


FIGURE 23. Example of partial diagram (reference 5.7.10).

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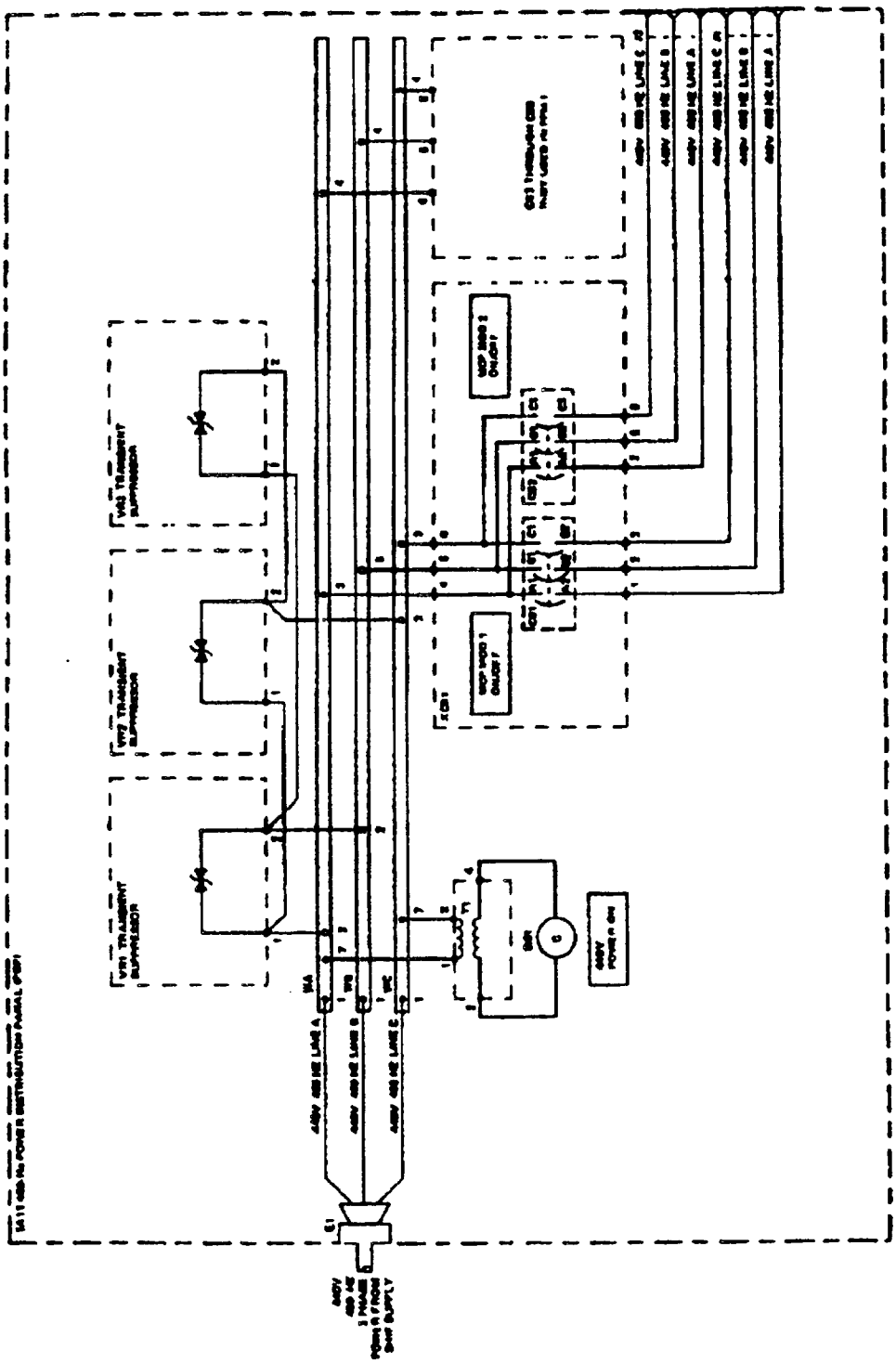


FIGURE 24. Example of power distribution diagram (reference 5.7.13).

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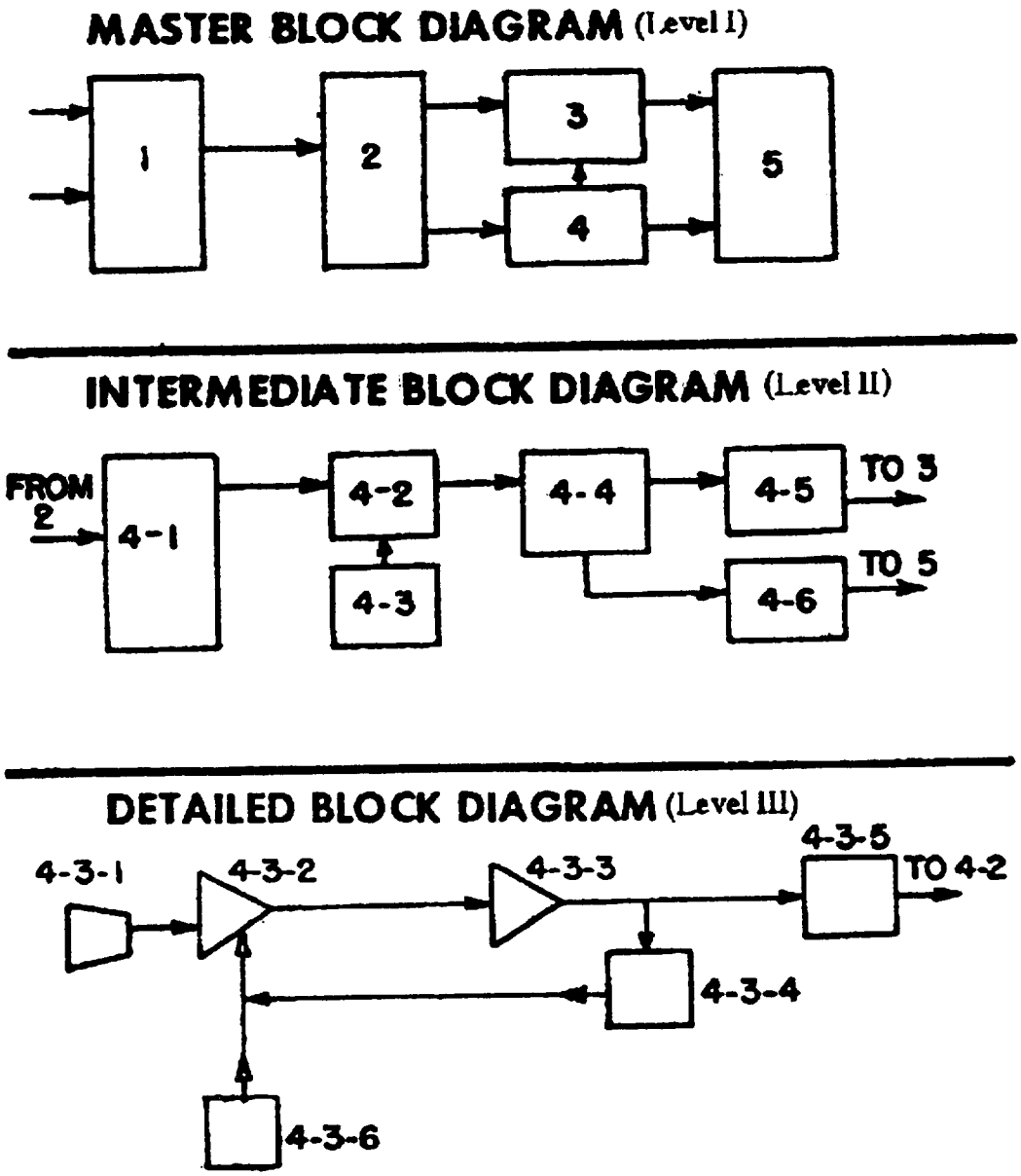


FIGURE 25. Example of pyramid diagram (reference 5.7.14).

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CABLE	FROM			TO		
	ASSEMBLY	JACK	CABLE PLUG	ASSEMBLY	JACK	CABLE PLUG
W101	CONVERTER	J1	P1	RECEIVER	J3	P2
W102	CONVERTER	J5	P1	RECEIVER	J1	P2
W103	CONVERTER	J7	P1	TRANSMITTER	J2	P2
W104	CONVERTER	J8	P1	RECEIVER	J2	P2
W106	CONVERTER	J6	P1	TRANSMITTER	J3	P2
W108	CONVERTER	J3	P1	TRANSMITTER	J4	P2
W107	CONVERTER	J4	P1	TRANSMITTER	J1	P2

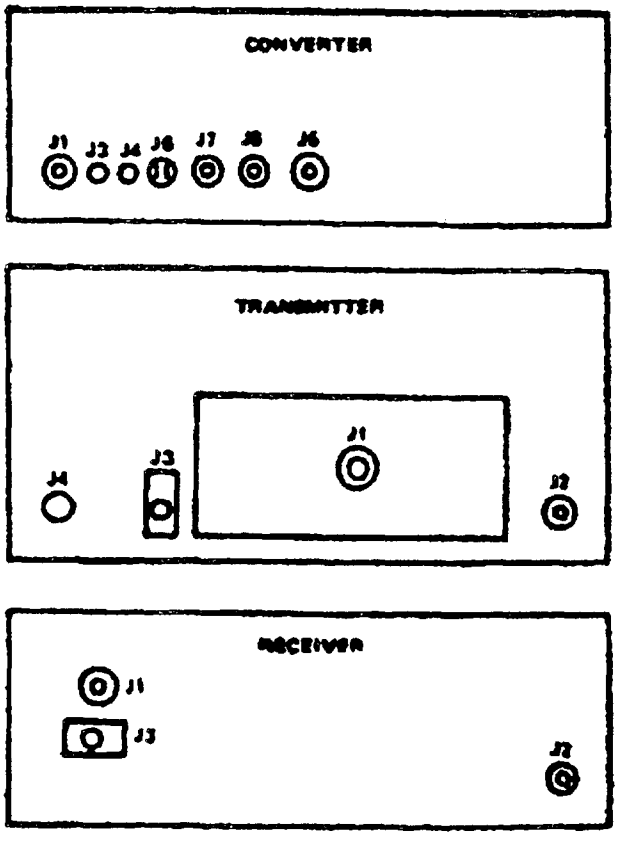


FIGURE 26. Example of cable diagram illustration and table (reference 5.8.1, 5.8.1.1, 5.8.1.2).

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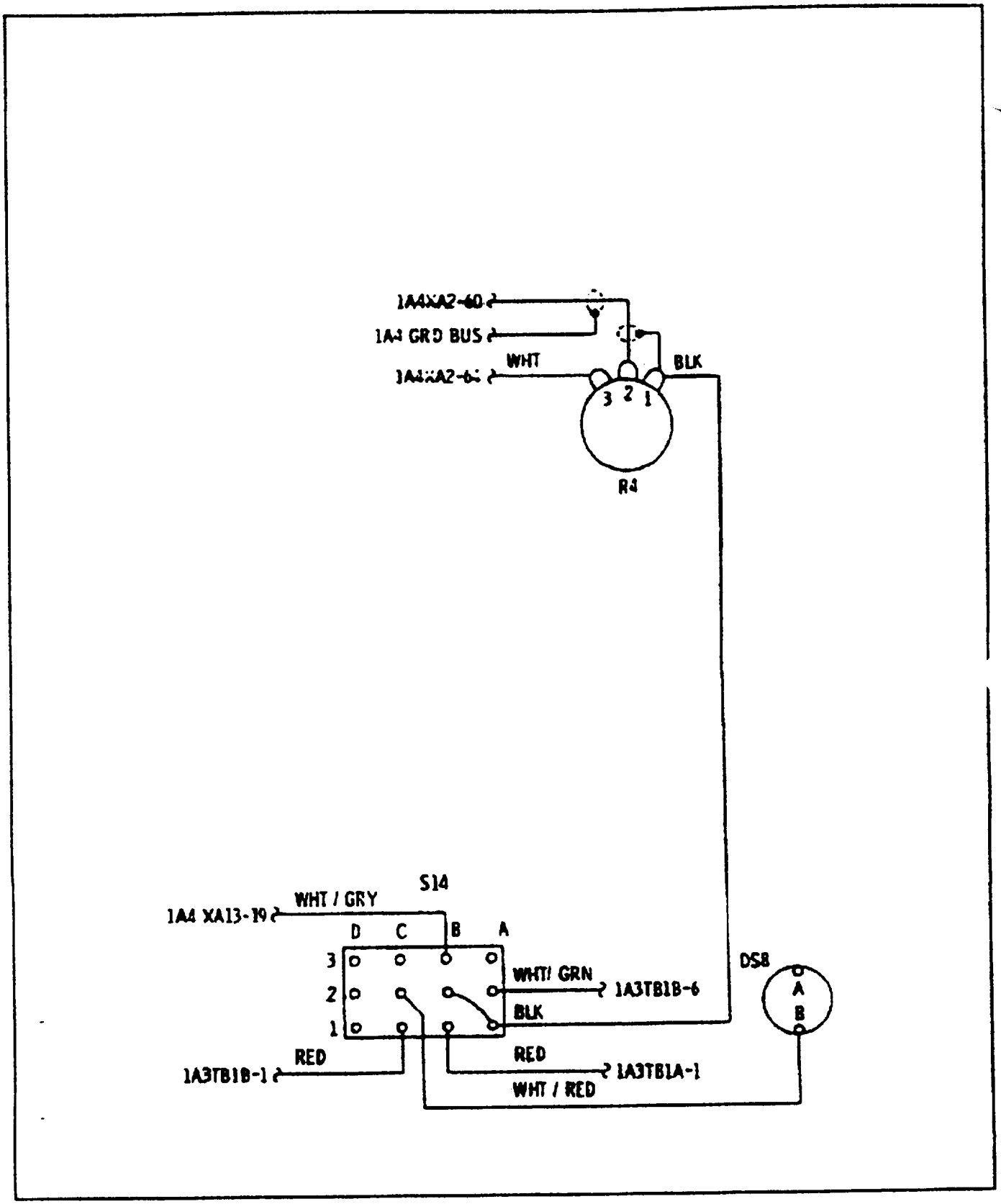


FIGURE 27. Example of wiring diagram (reference 5.8.2).

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FROM	SHD GRP	WIRE ID	TO	SHD GRP	ROUTE CODE
A17(Cont)					
J2-A		381	A17 J1-A		
J2-B		382	A17 J1-B		
J2-C		383	A17 J1-C		
J2-D		384	A17 J1-D		
S1	FS	385	A39	FT	
S1-1		351	A17 TB1-1		
S1-2		BLK385	A39		
S1-3		353	A17 TB1-3		
S1-4		352	A17 TB1-2		
S1-5		RED385	A39		
S1-6		354	A17 TB1-4		
S2-1		368	A17 S3-1		
S2-1		376	A17 TB2-1		
S2-2		355	A17 TB1-5		
S3-1		368	A17 S2-1		
S3-1		369	A17 S4-1		
S3-2		356	A17 TB1-6		
S4-1		369	A17 S3-1		
S4-1		370	A17 S5-1		
S4-2		357	A17 TB1-7		
S5-1		370	A17 S4-1		
S5-1		371	A17 S6-1		
S5-2		358	A17 TB1-8		
S6-1		371	A17 S5-1		
S6-1		372	A17 S7-1		
S6-2		359	A17 TB1-9		
S7-1		372	A17 S6-1		
S7-1		373	A17 S8-1		
S7-2		360	A17 TB1-10		
S8-1		373	A17 S7-1		
S8-1		374	A17 S9-4		
S8-2		361	A17 TB2-4		
S9-1		362	A17 TB2-5		
S9-2		363	A17 TB2-6		
S9-3		364	A17 TB2-7		
S9-4		374	A17 S8-1		
S9-4		375	A17 S10-4		
S10-1		365	A17 TB2-8		
S10-2		366	A17 TB2-9		
S10-3		367	A17 TB2-10		
S10-4		375	A17 S9-4		
TB1		386	TB12		A-D
TB1-1		351	A17 S1-1		
TB1-1		386A	TB12-1		A-D
TB1-2		352	A17 S1-4		
TB1-2		386B	TB12-2		A-D
TB1-3		353	A17 S1-3		
TB1-3		386C	TB12-3		A-D
TB1-4		354	A17 S1-6		
TB1-4		386D	TB13-4		A-D
TB1-5		355	A17 S2-2		
TB1-5		386E	TB12-5		A-D
TB1-6		356	A17 S3-2		
TB1-6		386F	TB12-6		A-D
TB1-7		357	A17 S4-2		
TB1-7		386G	TB12-7		A-D
TB1-8		358	A17 S5-2		

FROM	SHD GRP	WIRE ID	TO	SHD GRP	ROUTE CODE
A17(Cont)					
TB1-8		386H	TB12-8		A-D
TB1-9		359	A17 S7-2		
TB1-9		386J	TB12-9		A-D
TB1-10		360	A17 S7-2		
TB1-10		386K	TB12-10		A-D
TB2		367	TB13		A-D
TB2-1		376	A17 S2-1		
TB2-1		387A	TB13-2		A-D
TB2-2		388	A20 1		
TB2-2		387B	TB13-4		A-D
TB2-3		389	A20 2		
TB2-3		387C	TB13-6		A-D
TB2-4		361	A17 S8-2		
TB2-4		387D	TB12-11		A-D
TB2-5		362	A17 S9-1		
TB2-5		387E	TB12-12		A-D
TB2-6		363	A17 S9-2		
TB2-6		387F	TB12-13		A-D
TB2-7		364	A17 S9-3		
TB2-7		387G	TB12-14		A-D
TB2-8		365	A17 S10-1		
TB2-8		387H	TB13-12		A-D
TB2-9		366	A17 S10-2		
TB2-9		387J	TB13-13		A-D
TB2-10		367	A17 S10-3		
TB2-10		387K	TB13-14		A-D
A18					
E1		705	A18 STP	FU	
STP	FU	705	A18 E1		
J1-A		416	P10-A		
J1-A		420	A18 J2-A		
J1-B		417	P10-B		
J1-B		421	A18 J2-B		
J1-C		418	P10-C		
J1-C		422	A18 J2-C		
J1-D		419	P10-D		
J1-D		423	A18 J2-D		
J2-A		420	A18 J1-A		
J2-B		421	A18 J1-B		
J2-C		422	A18 J1-C		
J2-D		423	A18 J1-D		
S1	FU	424	A40	FV	
S1-1		390	A18 TB1-1		
S1-2		BLK424	A40		
S1-3		392	A18 TB1-3		
S1-4		391	A18 TB1-2		
S1-5		RED424	A40		
S1-6		393	A18 TB1-4		
S2-1		407	A18 S3-1		
S2-1		415	A18 TB2-1		
S2-2		394	A18 TB1-5		
S3-1		407	A18 S2-1		
S3-1		408	A18 S4-1		
S3-2		395	A18 TB1-6		
S4-1		408	A18 S3-1		
S4-1		409	A18 S5-1		
S4-2		396	A18 TB1-7		

FIGURE 28. Example of wire list (reference 5.8.3).

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